Resampling triggers, common threshold $0.5N$

Table 1: Linear Gaussian SSM: state RMSE (lower is better), common threshold $0.5N$. Resample when $E_2 = 1/\sum_i w_i^2$, $E_H = \exp(-\sum_i w_i \log w_i)$ or $E_\infty = 1/\max_i w_i$ falls below $0.5N$. $E_\infty \le E_2 \le E_H$, so E_∞ resamples most often and E_H least. $CV^2 > 1$ is the same rule as $E_2 < 0.5N$. 50 steps; 100 filter runs over 10 observation sequences (10 for OT and Gumbel).

	E_2				E_H				E_∞			
Method	32	128	512	2048	32	128	512	2048	32	128	512	2048
DSR (τ=0.1, paper)	0.52	0.506	0.505	0.504	0.523	**0.506**	0.504	0.503	**0.518**	0.507	**0.504**	**0.502**
DSR (τ=0.1, unsorted)	0.524	0.51	0.509	0.507	0.524	**0.509**	**0.506**	**0.504**	**0.522**	0.509	0.509	0.508
DSR (τ=0.5)	0.523	0.509	0.509	0.507	0.523	**0.508**	**0.506**	**0.504**	**0.523**	0.509	0.509	0.508
DSR (τ=2.5/N)†	0.521	**0.507**	0.504	0.503	0.524	0.507	0.504	0.503	**0.518**	0.507	**0.504**	**0.503**
Systematic	0.52	**0.506**	0.505	**0.503**	0.525$^-$	0.506	**0.504**	0.503	**0.516**$^+$	0.507	0.505	0.503
Multinomial	0.521	0.507	**0.504**	0.503	0.522	0.508	0.504	**0.503**	**0.521**	**0.506**	0.505	0.503
Systematic, stop-gradient (PyDPF)	0.521	0.506	0.505	0.503	0.525$^-$	0.508$^-$	**0.504**	**0.503**	**0.518**$^+$	**0.505**	0.505	0.503
Soft (α=0.5)	0.523	0.508	0.505	**0.503**	0.525	**0.507**	0.504	0.503	**0.523**	0.507	**0.503**	0.503
OT (PyDPF)	0.53	0.515	–	–	0.533	0.512	–	–	**0.528**	**0.51**$^+$	–	–
OT (PFJAX)	0.529	0.515	–	–	0.532	0.512	–	–	**0.527**	**0.51**$^+$	–	–
OT (FilterFlow)	0.53	0.515	–	–	0.535	0.512	–	–	**0.528**	**0.509**$^+$	–	–
Eigen moment matching	0.52	0.507	0.505	0.503	0.522	0.507	0.505	**0.503**	**0.519**	**0.506**$^+$	**0.505**	**0.503**
Gaussian moment matching	0.521	**0.507**	0.504	0.503	0.524	0.508	0.504	0.503	**0.521**	0.508	**0.504**	**0.503**$^+$
Gumbel-softmax	0.526	0.514	–	–	0.532	0.515	–	–	**0.525**	**0.512**	–	–
Variational (KL)	**0.519**	**0.507**	0.504	**0.503**	0.523$^-$	0.507	0.505	0.503	0.519	0.508	**0.504**	0.503

†Local diagnostic, not from the paper: τ rescaled so that τN equals the paper's value at N=25. Cells are means; $^+/^-$: paired difference from E_2 on the same trials significantly better/worse (95% CI over observation sequences excludes 0). Bold marks the best trigger for that method and N. A dash means not run (dense methods stop at N=128).

Table 2: Linear Gaussian SSM: posterior-mean RMSE vs. Kalman (lower is better), common threshold $0.5N$. Resample when $E_2 = 1/\sum_i w_i^2$, $E_H = \exp(-\sum_i w_i \log w_i)$ or $E_\infty = 1/\max_i w_i$ falls below $0.5N$. $E_\infty \le E_2 \le E_H$, so E_∞ resamples most often and E_H least. $CV^2 > 1$ is the same rule as $E_2 < 0.5N$. 50 steps; 100 filter runs over 10 observation sequences (10 for OT and Gumbel).

	E_2				E_H				E_∞			
Method	32	128	512	2048	32	128	512	2048	32	128	512	2048
DSR (τ=0.1, paper)	0.138	**0.0743**	**0.0442**	0.0339	0.146	0.078	0.0454	**0.0326**	**0.136**	0.0767	0.053	0.0438
DSR (τ=0.1, unsorted)	**0.151**	0.1	0.0803	0.0745	0.152	**0.1**	**0.0768**	**0.0703**	0.162	0.118	0.103	0.0989
DSR (τ=0.5)	**0.153**	**0.101**	0.0797	0.0732	0.155	0.102	**0.0767**	**0.069**	0.165	0.119	0.102	0.0972
DSR (τ=2.5/N)†	0.136	0.0668	0.0326	0.0171	0.145	0.0718	0.0347	0.0176	**0.133**	**0.0641**	**0.0319**	**0.0156**
Systematic	0.132	0.0675	0.0324	0.017	0.14	0.0717	0.0342	0.0177	**0.125**	**0.0666**	**0.0315**	**0.0159**
Multinomial	0.134	0.0692	**0.0336**	0.0177	0.143	0.0738	0.036	0.0177	**0.132**	**0.0672**	0.0339	**0.0161**
Systematic, stop-gradient (PyDPF)	0.131	0.0678	0.0329	0.0169	0.142	0.0711	0.035	0.0176	**0.126**	**0.0631**	**0.0317**	**0.0158**
Soft (α=0.5)	0.145	0.0754	0.0357	0.0182	0.151	0.0773	0.0392	0.0192	**0.138**	**0.0712**	**0.0343**	**0.0177**
OT (PyDPF)	0.127	0.067	–	–	0.135	0.0666	–	–	**0.121**	**0.0623**	–	–
OT (PFJAX)	0.127	0.0672	–	–	0.132	0.0666	–	–	**0.122**	**0.0624**	–	–
OT (FilterFlow)	0.128	0.0679	–	–	0.136	0.068	–	–	**0.123**	**0.0638**	–	–
Eigen moment matching	0.131	0.0678	0.0321	0.0166	0.138	0.0712	0.0339	0.0175	**0.123**	**0.065**	**0.0304**	**0.0156**
Gaussian moment matching	0.133	0.0707	0.0335	0.0172	0.143	0.0734	0.0358	0.0173	**0.133**	**0.0684**	**0.033**	**0.0164**
Gumbel-softmax	**0.133**	**0.0793**	–	–	0.152	0.0822	–	–	0.14	0.0831	–	–
Variational (KL)	0.132	0.067	**0.0334**	**0.0181**	0.14	0.0712	0.0348	0.0184	**0.126**	**0.0654**	0.0341	0.0186

†Local diagnostic, not from the paper: τ rescaled so that τN equals the paper's value at N=25. Cells are means; $^+/^-$: paired difference from E_2 on the same trials significantly better/worse (95% CI over observation sequences excludes 0). Bold marks the best trigger for that method and N. A dash means not run (dense methods stop at N=128).

Table 3: Linear Gaussian SSM: log-likelihood error vs. Kalman (closer to 0 is better), common threshold $0.5N$. Resample when $E_2 = 1/\sum_i w_i^2$, $E_H = \exp(-\sum_i w_i \log w_i)$ or $E_\infty = 1/\max_i w_i$ falls below $0.5N$. $E_\infty \le E_2 \le E_H$, so E_∞ resamples most often and E_H least. $CV^2 > 1$ is the same rule as $E_2 < 0.5N$. 50 steps; 100 filter runs over 10 observation sequences (10 for OT and Gumbel).

	E_2				E_H				E_∞			
Method	32	128	512	2048	32	128	512	2048	32	128	512	2048
DSR (τ=0.1, paper)	−0.847	**−0.19**	**−0.0486**	**0.0111**	**−0.818**	−0.265	−0.0568	−0.0169	−0.823	−0.261	−0.128	−0.0882
DSR (τ=0.1, unsorted)	**−0.854**	**−0.265**	−0.195	−0.117	−0.937	−0.316	**−0.147**	**−0.109**	−1.17	−0.532	−0.413	−0.365
DSR (τ=0.5)	**−0.887**	**−0.235**	−0.195	−0.114	−0.955	−0.326	**−0.155**	**−0.099**	−1.17	−0.538	−0.403	−0.354
DSR (τ=2.5/N)†	−0.839	**−0.205**	−0.0415	0.00311	−0.856	−0.244	−0.0464	$\mathbf{5.5 \times 10^{-4}}$	**−0.748**	−0.206	**−0.034**	0.0144
Systematic	−0.829	−0.186	−0.0508	0.0258	−0.78	−0.247	**−0.0218**	**0.00229**	**−0.602**	**−0.154**	−0.0414	0.0166
Multinomial	−0.903	−0.207	−0.0361	0.0103	−0.833	−0.2	**−0.0253**	**−0.00291**	**−0.824**	**−0.167**	−0.0262	−0.0126
Systematic, stop-gradient (PyDPF)	−0.756	−0.171	**−0.0196**	**−0.00821**	−0.783	−0.25	−0.0346	−0.011	**−0.668**	**−0.163**	−0.0266	0.0162
Soft (α=0.5)	−0.93	−0.333	**−0.0328**	**−0.0124**	−1.01	−0.268	−0.0378	−0.0195	**−0.692**	**−0.221**	−0.112	−0.0135
OT (PyDPF)	−1.32	−0.277	–	–	−1.33	**−0.262**	–	–	**−1.17**	−0.402	–	–
OT (PFJAX)	−1.28	−0.276	–	–	−1.25	**−0.261**	–	–	**−1.14**	−0.396	–	–
OT (FilterFlow)	−1.33	**−0.269**	–	–	−1.36	−0.289	–	–	**−1.17**	−0.4	–	–
Eigen moment matching	−0.852	−0.23	−0.0483	$\mathbf{6.98 \times 10^{-4}}$	−0.853	−0.212	−0.0437	−0.00292	**−0.716**	**−0.187**	**−0.0373**	0.00625
Gaussian moment matching	**−0.825**	−0.143	−0.064	0.00867	−0.867	−0.185	**−0.0432**	**−0.00237**	−0.83	**−0.0896**	−0.0567	0.0129
Gumbel-softmax	−1.33	**−0.236**	–	–	−1.57	−0.345	–	–	**−1.2**	−0.429	–	–
Variational (KL)	−0.827	**−0.145**	−0.041	0.00757	−0.781	−0.238	**−0.0357**	**0.0026**	**−0.706**	−0.209	−0.0775	−0.0347

†Local diagnostic, not from the paper: τ rescaled so that τN equals the paper's value at N=25. Cells are means; $^+/^-$: paired difference from E_2 on the same trials significantly better/worse (95% CI over observation sequences excludes 0). Bold marks the best trigger for that method and N. A dash means not run (dense methods stop at N=128).

Table 4: Linear Gaussian SSM: 95% coverage (closer to 0.95 is better), common threshold $0.5N$. Resample when $E_2 = 1/\sum_i w_i^2$, $E_H = \exp(-\sum_i w_i \log w_i)$ or $E_\infty = 1/\max_i w_i$ falls below $0.5N$. $E_\infty \le E_2 \le E_H$, so E_∞ resamples most often and E_H least. $CV^2 > 1$ is the same rule as $E_2 < 0.5N$. 50 steps; 100 filter runs over 10 observation sequences (10 for OT and Gumbel).

	E_2				E_H				E_∞			
Method	32	128	512	2048	32	128	512	2048	32	128	512	2048
DSR (τ=0.1, paper)	**0.903**	0.923	0.926	0.928	0.902	**0.925**	**0.926**	**0.929**	0.901	0.921	0.926	0.928
DSR (τ=0.1, unsorted)	0.895	0.913	0.916	0.917	**0.897**	**0.914**	**0.919**	**0.922**	0.886^-	0.904	0.911	0.913
DSR (τ=0.5)	**0.894**	0.912	0.916	0.917	0.893	**0.914**	**0.919**	**0.922**	0.883^-	0.904	0.91	0.914
DSR (τ=2.5/N)†	**0.904**	0.929	0.931	**0.933**	0.904	**0.929**	0.931	**0.933**	**0.904**	0.927	**0.931**	0.933
Systematic	0.912	**0.929**	**0.931**	**0.933**	0.907	0.927	0.931	0.933	**0.914**	0.928	0.931	0.933
Multinomial	**0.911**	0.928	0.931	0.933	0.908	0.927	**0.932**	**0.933**	0.911	**0.928**	0.931	0.932
Systematic, stop-gradient (PyDPF)	0.909	**0.931**	0.931	0.932	0.905	0.928	**0.931**	**0.934**	**0.914**	0.93	**0.931**	0.933
Soft (α=0.5)	**0.908**	**0.927**	0.931	**0.932**	0.899	0.926	**0.931**	0.932	0.904	0.926	0.929	0.932
OT (PyDPF)	**0.916**	**0.928**	–	–	0.906	**0.928**	–	–	0.914	0.926	–	–
OT (PFJAX)	**0.918**	**0.93**	–	–	0.908	0.928	–	–	0.914	0.926	–	–
OT (FilterFlow)	**0.916**	0.924	–	–	0.9	0.924	–	–	0.914	**0.926**	–	–
Eigen moment matching	**0.914**	**0.929**	0.931	0.932	0.909	**0.929**	0.931	0.932	0.914	0.928	**0.931**	**0.933**
Gaussian moment matching	**0.911**	**0.929**	0.931	**0.934**	0.907	0.929	0.931	0.934	0.91	0.926^-	**0.932**	0.933
Gumbel-softmax	**0.902**	**0.928**	–	–	0.894	0.924	–	–	0.892	0.912	–	–
Variational (KL)	0.909	**0.929**	0.93	0.932	0.909	0.926^-	0.931	0.932	**0.913**	0.927	**0.932**	**0.933**

†Local diagnostic, not from the paper: τ rescaled so that τN equals the paper's value at N=25. Cells are means; $^+/^-$: paired difference from E_2 on the same trials significantly better/worse (95% CI over observation sequences excludes 0). Bold marks the best trigger for that method and N. A dash means not run (dense methods stop at N=128).

Table 5: Linear Gaussian SSM: predictive NLL (lower is better), common threshold $0.5N$. Resample when $E_2 = 1/\sum_i w_i^2$, $E_H = \exp(-\sum_i w_i \log w_i)$ or $E_\infty = 1/\max_i w_i$ falls below $0.5N$. $E_\infty \le E_2 \le E_H$, so E_∞ resamples most often and E_H least. $CV^2 > 1$ is the same rule as $E_2 < 0.5N$. 50 steps; 100 filter runs over 10 observation sequences (10 for OT and Gumbel).

	E_2				E_H				E_∞			
Method	32	128	512	2048	32	128	512	2048	32	128	512	2048
DSR (τ=0.1, paper)	1.51	**1.5**	**1.49**	**1.49**	**1.51**	1.5	1.49	1.49	1.51	1.5	1.5	1.49
DSR (τ=0.1, unsorted)	**1.51**	**1.5**	1.5	1.5	1.51	1.5	**1.5**	**1.5**	1.52	1.5	1.5	1.5
DSR (τ=0.5)	**1.51**	**1.5**	1.5	1.5	1.51	1.5	**1.5**	**1.49**	1.52	1.5	1.5	1.5
DSR (τ=2.5/N)†	1.51	**1.5**	1.49	1.49	1.51	1.5	1.49	1.49	**1.51**	1.5	**1.49**	**1.49**
Systematic	1.51	1.5	1.49	**1.49**	1.51	1.5	**1.49**	1.49	$\mathbf{1.5}^+$	**1.5**	1.49	1.49
Multinomial	1.51	1.5	1.49	**1.49**	1.51	1.5	**1.49**	1.49	**1.51**	**1.5**	1.49	1.49
Systematic, stop-gradient (PyDPF)	1.51	1.5	**1.49**	1.49	1.51	1.5	1.49	1.49	**1.51**	**1.5**	1.49	**1.49**
Soft (α=0.5)	1.51	1.5	**1.49**	**1.49**	1.51	1.5	1.49	1.49	**1.51**	**1.5**	1.5	1.49
OT (PyDPF)	1.52	1.5	–	–	1.52	**1.5**	–	–	**1.52**	1.5	–	–
OT (PFJAX)	1.52	1.5	–	–	1.52	**1.5**	–	–	**1.52**	1.5	–	–
OT (FilterFlow)	1.52	**1.5**	–	–	1.52	1.5	–	–	**1.52**	1.5	–	–
Eigen moment matching	1.51	1.5	1.49	1.49	1.51	1.5	1.49	1.49	$\mathbf{1.51}^+$	**1.5**	**1.49**	**1.49**
Gaussian moment matching	**1.51**	1.5	1.49	1.49	1.51	1.5	**1.49**	1.49	1.51	**1.49**	1.49	**1.49**
Gumbel-softmax	1.52	**1.5**	–	–	1.52	1.5	–	–	**1.52**	1.5	–	–
Variational (KL)	1.51	**1.5**	1.49	**1.49**	1.51	1.5	**1.49**	1.49	**1.51**	1.5^-	1.49	1.49

†Local diagnostic, not from the paper: τ rescaled so that τN equals the paper's value at N=25. Cells are means; $^+/^-$: paired difference from E_2 on the same trials significantly better/worse (95% CI over observation sequences excludes 0). Bold marks the best trigger for that method and N. A dash means not run (dense methods stop at N=128).

Table 6: Linear Gaussian SSM: resampling steps out of 50, common threshold $0.5N$. Resample when $E_2 = 1/\sum_i w_i^2$, $E_H = \exp(-\sum_i w_i \log w_i)$ or $E_\infty = 1/\max_i w_i$ falls below $0.5N$. $E_\infty \le E_2 \le E_H$, so E_∞ resamples most often and E_H least. $CV^2 > 1$ is the same rule as $E_2 < 0.5N$. 50 steps; 100 filter runs over 10 observation sequences (10 for OT and Gumbel).

	E_2				E_H				E_∞			
Method	32	128	512	2048	32	128	512	2048	32	128	512	2048
DSR (τ=0.1, paper)	15.5	16.2	16.3	16.2	12.2	12.7	12.9	12.9	26.2	26.8	26.9	26.9
DSR (τ=0.1, unsorted)	15.1	15.6	15.8	15.7	11.9	12.2	12.3	12.3	25.7	26.2	26.3	26.3
DSR (τ=0.5)	15	15.6	15.8	15.7	11.8	12.2	12.3	12.3	25.6	26.2	26.3	26.3
DSR (τ=2.5/N)†	15.6	16.5	16.5	16.4	12.2	13.1	13.2	13.2	26.3	27.4	27.4	27.7
Systematic	16.1	16.5	16.5	16.5	12.4	13	13.2	13.2	26.8	27.3	27.5	27.6
Multinomial	15.8	16.5	16.5	16.5	12.5	13	13.2	13.2	26.8	27.3	27.4	27.6
Systematic, stop-gradient (PyDPF)	15.9	16.4	16.6	16.4	12.6	13.1	13.3	13.3	26.7	27.4	27.5	27.6
Soft (α=0.5)	22.7	23.3	23.6	23.6	17.6	18.1	18.2	18.5	37.7	39.5	40.5	41.1
OT (PyDPF)	15.9	16.6	–	–	12.6	13.1	–	–	26.7	27.5	–	–
OT (PFJAX)	15.8	16.6	–	–	12.7	13.1	–	–	26.7	27.5	–	–
OT (FilterFlow)	15.9	16.6	–	–	12.6	13.1	–	–	26.7	27.4	–	–
Eigen moment matching	16	16.4	16.5	16.5	12.7	13	13.2	13.2	26.7	27.2	27.4	27.6
Gaussian moment matching	16	16.4	16.6	16.5	12.5	13	13.2	13.2	26.6	27.3	27.5	27.6
Gumbel-softmax	15.8	16.1	–	–	12.5	12.9	–	–	25.8	26.8	–	–
Variational (KL)	15.8	16.5	16.6	16.5	12.5	13	13.2	13.2	26.7	27.2	27.4	27.4

†Local diagnostic, not from the paper: τ rescaled so that τN equals the paper's value at N=25. Cells are means; $^+/^-$: paired difference from E_2 on the same trials significantly better/worse (95% CI over observation sequences excludes 0). Bold marks the best trigger for that method and N. A dash means not run (dense methods stop at N=128).

Table 7: UNGM: state RMSE (lower is better), common threshold $0.5N$. Resample when $E_2 = 1/\sum_i w_i^2$, $E_H = \exp(-\sum_i w_i \log w_i)$ or $E_\infty = 1/\max_i w_i$ falls below $0.5N$. $E_\infty \le E_2 \le E_H$, so E_∞ resamples most often and E_H least. $CV^2 > 1$ is the same rule as $E_2 < 0.5N$. 50 steps; 100 filter runs over 10 observation sequences (10 for OT and Gumbel).

	E_2				E_H				E_∞			
Method	32	128	512	2048	32	128	512	2048	32	128	512	2048
DSR (τ=0.1, paper)	6.22	5.15	4.79	4.78	6.35	5.2	**4.69**	**4.67**	**6.05**	**5.13**	4.83	4.68
DSR (τ=0.1, unsorted)	**7.52**	7.63	8.64	8.68	7.61	**7.51**	**8.09**	**8.44**	8.15^-	8.52^-	9.05	9.44
DSR (τ=0.5)	8.64	8.52	8.81	8.88	**8.55**	**8.15**	**8.22**	**8.14**	8.66	8.85	9.15	9.19
DSR (τ=2.5/N)†	6.02	4.98	**4.56**	**4.52**	6.25^-	5.04	4.59	4.52	**5.91**	**4.91**	4.58	4.53
Systematic	**6.01**	4.91	4.49	**4.48**	6.1	**4.86**	4.56	4.53	6.04	5	**4.47**	4.5
Multinomial	5.97	4.92	4.57	4.52	6.05	**4.77**	4.56	4.48	**5.72**	4.85	**4.51**	**4.46**
Systematic, stop-gradient (PyDPF)	**5.83**	4.88	4.51	4.54	5.92	**4.79**	4.56	4.52	5.94	4.86	**4.5**	**4.49**
Soft (α=0.5)	**5.42**	**4.83**	4.54	4.44	5.56	4.83	4.55	**4.44**	5.7	4.88	**4.51**	4.45
OT (PyDPF)	5.6	5.42	–	–	5.84	5.32	–	–	**5.52**	**5.24**	–	–
OT (PFJAX)	**6.1**	5.33	–	–	6.2	5.3	–	–	6.27	**5.18**	–	–
OT (FilterFlow)	6.01	5.48	–	–	6.22	5.34^+	–	–	**5.53**	**5.12**	–	–
Eigen moment matching	**5.99**	5.17	4.98	4.83	6.22^-	**5.1**	**4.95**	**4.74**	5.99	5.28	5.12^-	4.97
Gaussian moment matching	6.02	**4.94**	4.8	**4.66**	6.18	5.02	4.77	4.67	**5.82**	4.95	**4.76**	4.66
Gumbel-softmax	6.51	5.27	–	–	6.3	5.4	–	–	$\mathbf{5.86}^+$	**5.06**	–	–
Variational (KL)	5.79	4.83	**4.56**	4.48	5.89	4.86	4.61	4.53	**5.74**	**4.72**	4.57	**4.43**

†Local diagnostic, not from the paper: τ rescaled so that τN equals the paper's value at N=25. Cells are means; $^+/^-$: paired difference from E_2 on the same trials significantly better/worse (95% CI over observation sequences excludes 0). Bold marks the best trigger for that method and N. A dash means not run (dense methods stop at N=128).

Table 8: UNGM: 95% coverage (closer to 0.95 is better), common threshold $0.5N$. Resample when $E_2 = 1/\sum_i w_i^2$, $E_H = \exp(-\sum_i w_i \log w_i)$ or $E_\infty = 1/\max_i w_i$ falls below $0.5N$. $E_\infty \le E_2 \le E_H$, so E_∞ resamples most often and E_H least. $CV^2 > 1$ is the same rule as $E_2 < 0.5N$. 50 steps; 100 filter runs over 10 observation sequences (10 for OT and Gumbel).

	E_2				E_H				E_∞			
Method	32	128	512	2048	32	128	512	2048	32	128	512	2048
DSR (τ=0.1, paper)	0.813	0.926	0.955	**0.961**	0.808	0.925	0.958	0.964	**0.82**	**0.927**	**0.954**	0.963
DSR (τ=0.1, unsorted)	0.737	0.784	0.763	0.766	**0.74**	**0.788**	**0.79**	**0.778**	0.706^-	0.738^-	0.727^-	0.724^-
DSR (τ=0.5)	0.672	0.738	0.752	0.768	**0.684**	**0.75**	**0.778**	**0.8**	0.665	0.717	0.728	0.747
DSR (τ=2.5/N)†	0.822	0.929	0.963	0.971	0.814^-	0.929	**0.963**	0.971	**0.825**	**0.933**	0.963	**0.97**
Systematic	0.821	0.93	0.966	0.971	0.818	**0.931**	**0.964**	**0.971**	**0.823**	0.929	0.967	0.971
Multinomial	0.818	0.928	0.964	0.972	0.814	0.934	**0.963**	0.972	**0.829**	**0.936**	0.964	**0.971**
Systematic, stop-gradient (PyDPF)	**0.828**	0.932	0.966	0.972	0.823	**0.936**	**0.964**	**0.971**	0.819	0.934	0.966	0.972
Soft (α=0.5)	0.797	**0.928**	**0.962**	0.97	**0.798**	0.921	**0.962**	0.97	0.793	0.924	0.965	**0.969**
OT (PyDPF)	0.814	0.914	–	–	0.816	0.914	–	–	**0.842**	**0.922**	–	–
OT (PFJAX)	0.8	**0.918**	–	–	0.802	0.916	–	–	**0.814**	**0.918**	–	–
OT (FilterFlow)	0.806	0.908	–	–	0.804	0.914	–	–	**0.838**	**0.924**	–	–
Eigen moment matching	**0.828**	0.93	**0.953**	0.963	0.816^-	**0.932**	0.954	0.965	0.826	0.925	0.946	**0.959**
Gaussian moment matching	0.824	0.933	0.962	0.97	0.82	0.932	**0.961**	**0.969**	**0.837**	**0.936**	0.963	0.97
Gumbel-softmax	0.82	0.918	–	–	0.804	0.912	–	–	**0.826**	**0.932**	–	–
Variational (KL)	0.825	0.93	0.963	**0.971**	0.825	0.931	**0.962**	0.971	**0.827**	**0.934**	0.964	0.971

†Local diagnostic, not from the paper: τ rescaled so that τN equals the paper's value at N=25. Cells are means; $^+/^-$: paired difference from E_2 on the same trials significantly better/worse (95% CI over observation sequences excludes 0). Bold marks the best trigger for that method and N. A dash means not run (dense methods stop at N=128).

Table 9: UNGM: predictive NLL (lower is better), common threshold $0.5N$. Resample when $E_2 = 1/\sum_i w_i^2$, $E_H = \exp(-\sum_i w_i \log w_i)$ or $E_\infty = 1/\max_i w_i$ falls below $0.5N$. $E_\infty \le E_2 \le E_H$, so E_∞ resamples most often and E_H least. $CV^2 > 1$ is the same rule as $E_2 < 0.5N$. 50 steps; 100 filter runs over 10 observation sequences (10 for OT and Gumbel).

	E_2				E_H				E_∞			
Method	32	128	512	2048	32	128	512	2048	32	128	512	2048
DSR (τ=0.1, paper)	5.53	3.23	2.86	2.77	5.46	3.32	**2.85**	2.76	**5.24**	**3.21**	2.87	**2.72**
DSR (τ=0.1, unsorted)	**7.21**	6.09	6.2	5.57	7.28	**5.94**	**5.84**	**5.45**	7.92	6.37	6.22	5.61
DSR (τ=0.5)	9.33	7.06	6.21	5.46	9.33	**6.46**	**5.8**	**5.01**	**9.17**	6.94	6.21	5.34
DSR (τ=2.5/N)†	5.33	3.19	**2.8**	2.7	5.49	3.23	2.82	2.7	**5.03**	**3.15**	2.82	$\mathbf{2.7}^+$
Systematic	5.36	3.18	**2.78**	**2.7**	5.29	**3.15**	2.81	2.71	**5.28**	3.33	2.79	2.71
Multinomial	5.26	3.3	2.82	2.7	5.23	3.15	2.82	**2.7**	$\mathbf{4.88}^+$	**3.14**	**2.77**	2.7
Systematic, stop-gradient (PyDPF)	**4.99**	3.14	2.79	**2.7**	4.99	**3.13**	2.82	2.71	5.23	3.19	**2.78**	2.7
Soft (α=0.5)	**4.32**	**3.13**	2.81	2.71	4.37	3.18	2.83	2.71	4.47	3.17	**2.76**	**2.7**
OT (PyDPF)	**4.18**	**3.18**	–	–	4.73	3.19	–	–	4.21	3.19	–	–
OT (PFJAX)	**4.6**	3.19	–	–	5.21	3.2	–	–	5.03	**3.18**	–	–
OT (FilterFlow)	4.53	3.2	–	–	4.78	3.2	–	–	**4.18**	**2.96**	–	–
Eigen moment matching	4.98	**3.17**	**2.88**	2.77	5.08	3.25	2.89	**2.76**	**4.89**	3.2	2.91	2.77
Gaussian moment matching	5.06	3.13	2.82	**2.71**	5.18	3.17	2.84	2.71	**4.83**	**3.09**	**2.8**	2.71
Gumbel-softmax	4.82	3.16	–	–	5.07	3.19	–	–	**4.44**	**2.95**	–	–
Variational (KL)	4.98	**3.18**	2.83	2.71	5.05	3.19	2.83	**2.7**	**4.83**	3.18	**2.81**	2.7

†Local diagnostic, not from the paper: τ rescaled so that τN equals the paper's value at N=25. Cells are means; $^+/^-$: paired difference from E_2 on the same trials significantly better/worse (95% CI over observation sequences excludes 0). Bold marks the best trigger for that method and N. A dash means not run (dense methods stop at N=128).

Table 10: UNGM: resampling steps out of 50, common threshold $0.5N$. Resample when $E_2 = 1/\sum_i w_i^2$, $E_H = \exp(-\sum_i w_i \log w_i)$ or $E_\infty = 1/\max_i w_i$ falls below $0.5N$. $E_\infty \le E_2 \le E_H$, so E_∞ resamples most often and E_H least. $CV^2 > 1$ is the same rule as $E_2 < 0.5N$. 50 steps; 100 filter runs over 10 observation sequences (10 for OT and Gumbel).

	E_2				E_H				E_∞			
Method	32	128	512	2048	32	128	512	2048	32	128	512	2048
DSR (τ=0.1, paper)	37	37.4	37.7	37.8	35.2	35.6	35.7	35.8	41.1	41.7	41.9	41.8
DSR (τ=0.1, unsorted)	37.2	37.5	37.5	37.5	35.4	35.5	35.4	35.4	41.2	41.5	41.4	41.3
DSR (τ=0.5)	37.4	37.5	37.7	37.9	35.3	35.5	35.8	35.8	40.9	41.3	41.4	41.5
DSR (τ=2.5/N)†	37	37.3	37.3	37.2	35.2	35.5	35.4	35.4	41.2	41.6	41.7	41.7
Systematic	36.9	37.1	37.3	37.2	35	35.3	35.5	35.4	41.1	41.6	41.7	41.7
Multinomial	36.7	37.3	37.3	37.3	35	35.4	35.5	35.5	40.9	41.5	41.7	41.7
Systematic, stop-gradient (PyDPF)	36.8	37.2	37.3	37.3	35.1	35.4	35.5	35.5	41	41.6	41.7	41.6
Soft (α=0.5)	44.1	44.1	43.9	43.9	42.7	42.8	42.8	42.6	48.3	49	49.4	49.5
OT (PyDPF)	37	37.4	–	–	35.2	35.2	–	–	41.9	41.8	–	–
OT (PFJAX)	37.2	37.1	–	–	35.6	35.2	–	–	41.9	41.7	–	–
OT (FilterFlow)	37	37.6	–	–	35.6	35.2	–	–	41.6	41.6	–	–
Eigen moment matching	37.5	38.1	38.4	38.6	35.7	36.3	36.6	36.7	41.6	42	42.3	42.4
Gaussian moment matching	37.5	37.9	38.5	38.5	35.7	36.1	36.5	36.4	41.5	42	42.3	42.4
Gumbel-softmax	37.6	37.6	–	–	35.7	36	–	–	41.3	41.9	–	–
Variational (KL)	36.8	37.1	37.4	37.2	35	35.4	35.6	35.4	41.1	41.5	41.7	41.6

†Local diagnostic, not from the paper: τ rescaled so that τN equals the paper's value at N=25. Cells are means; $^+/^-$: paired difference from E_2 on the same trials significantly better/worse (95% CI over observation sequences excludes 0). Bold marks the best trigger for that method and N. A dash means not run (dense methods stop at N=128).

Table 11: Stochastic volatility: state RMSE (lower is better), common threshold $0.5N$. Resample when $E_2 = 1/\sum_i w_i^2$, $E_H = \exp(-\sum_i w_i \log w_i)$ or $E_\infty = 1/\max_i w_i$ falls below $0.5N$. $E_\infty \le E_2 \le E_H$, so E_∞ resamples most often and E_H least. $CV^2 > 1$ is the same rule as $E_2 < 0.5N$. 50 steps; 100 filter runs over 10 observation sequences (10 for OT and Gumbel).

	E_2				E_H				E_∞			
Method	32	128	512	2048	32	128	512	2048	32	128	512	2048
DSR (τ=0.1, paper)	0.425	0.413	0.412	0.411	0.43	0.416	0.416	0.416	**0.424**	**0.406**	**0.406**	**0.405**
DSR (τ=0.1, unsorted)	**0.431**	0.423	**0.418**	**0.418**	0.431	**0.418**	0.421	0.42	0.434	0.425	0.422	0.423
DSR (τ=0.5)	**0.431**	0.423	**0.418**	**0.418**	0.432	**0.419**	0.42	0.42	0.44	0.424	0.422	0.422
DSR (τ=2.5/N)†	0.425	0.415	0.415	0.414	0.431^-	0.417	0.415	0.414	**0.423**	**0.413**	**0.415**	**0.414**
Systematic	0.428	0.417	0.415	**0.414**	0.43	0.418	**0.414**	0.415^-	**0.427**	**0.417**	0.415	0.414
Multinomial	**0.43**	0.417	**0.415**	**0.414**	0.431	0.418	0.415	0.415	0.437	**0.417**	0.415	0.415
Systematic, stop-gradient (PyDPF)	**0.427**	0.416	**0.414**	0.414	0.429	0.417	0.414	0.414	0.429	**0.416**	0.415^-	**0.414**
Soft (α=0.5)	**0.429**	0.419	0.415	0.415	0.429	0.418	0.416	0.414	0.437	**0.417**	**0.415**	**0.414**
OT (PyDPF)	0.426	0.418	–	–	0.426	0.422	–	–	**0.426**	**0.417**	–	–
OT (PFJAX)	**0.426**	0.418	–	–	0.426	0.423	–	–	0.426	**0.418**	–	–
OT (FilterFlow)	0.427	0.417	–	–	**0.425**	0.422	–	–	0.426	**0.416**	–	–
Eigen moment matching	0.428	0.416	**0.415**	**0.414**	0.428	0.418	0.415	0.414	**0.427**	**0.415**	0.415	0.415
Gaussian moment matching	**0.426**	**0.416**	**0.415**	**0.414**	0.428	0.418	0.415	0.414	0.428	0.418	0.416	0.414
Gumbel-softmax	0.427	0.415	–	–	**0.423**	0.428	–	–	0.438	**0.414**	–	–
Variational (KL)	0.426	0.417	0.413	0.413	0.431^-	0.417	0.415^-	0.415	**0.42**	**0.405**	**0.404**	**0.401**

†Local diagnostic, not from the paper: τ rescaled so that τN equals the paper's value at N=25. Cells are means; $^+/^-$: paired difference from E_2 on the same trials significantly better/worse (95% CI over observation sequences excludes 0). Bold marks the best trigger for that method and N. A dash means not run (dense methods stop at N=128).

Table 12: Stochastic volatility: 95% coverage (closer to 0.95 is better), common threshold $0.5N$. Resample when $E_2 = 1/\sum_i w_i^2$, $E_H = \exp(-\sum_i w_i \log w_i)$ or $E_\infty = 1/\max_i w_i$ falls below $0.5N$. $E_\infty \le E_2 \le E_H$, so E_∞ resamples most often and E_H least. $CV^2 > 1$ is the same rule as $E_2 < 0.5N$. 50 steps; 100 filter runs over 10 observation sequences (10 for OT and Gumbel).

	E_2				E_H				E_∞			
Method	32	128	512	2048	32	128	512	2048	32	128	512	2048
DSR (τ=0.1, paper)	**0.926**	0.947	**0.954**	**0.952**	0.923	**0.95**	0.958	0.959	0.897	0.922	0.92	0.918
DSR (τ=0.1, unsorted)	0.886	0.896	0.905	0.904	**0.902**	**0.922**$^+$	**0.923**	**0.922**	0.822$^-$	0.824	0.818$^-$	0.802$^-$
DSR (τ=0.5)	0.88	0.893	0.907	0.908	**0.893**	**0.921**$^+$	**0.924**	**0.926**	0.812$^-$	0.826	0.819	0.811$^-$
DSR (τ=2.5/N)†	**0.931**	0.962	0.975	0.976	0.927	**0.96**	**0.973**	**0.975**	0.91	0.963	0.974	0.976
Systematic	**0.941**	0.966	0.974	0.976	0.937	**0.962**	**0.973**	**0.975**	0.937	0.964	0.974	0.976
Multinomial	**0.942**	0.961	0.974	**0.975**	0.935	**0.96**	**0.973**	0.976	0.928	0.964	0.974	0.975
Systematic, stop-gradient (PyDPF)	**0.942**	0.967	0.976	0.977	0.937	**0.963**	**0.973**	**0.976**	0.941	**0.963**	0.975	0.979$^-$
Soft (α=0.5)	**0.937**	**0.959**	**0.973**	**0.977**	0.933	0.961	0.974	0.977	0.934	0.963	0.973	**0.977**
OT (PyDPF)	0.94	0.97	–	–	**0.946**	**0.952**	–	–	**0.946**	0.964	–	–
OT (PFJAX)	0.938	0.97	–	–	0.944	**0.952**	–	–	**0.948**	0.966	–	–
OT (FilterFlow)	0.936	0.968	–	–	**0.946**	**0.95**	–	–	0.938	0.964	–	–
Eigen moment matching	0.937	0.966	0.975	0.978	0.937	**0.962**	**0.975**	**0.978**	**0.942**	0.968	0.975	0.978
Gaussian moment matching	0.939	0.965	0.977	0.978	0.935	**0.962**	0.973	**0.975**	**0.945**	0.963	**0.973**	0.977
Gumbel-softmax	0.916	**0.946**	–	–	**0.93**	0.938	–	–	0.89	0.908	–	–
Variational (KL)	0.938	**0.96**	0.972	0.973	0.935	0.961$^-$	**0.97**	**0.971**	**0.943**	0.971	0.977	0.974

†Local diagnostic, not from the paper: τ rescaled so that τN equals the paper's value at N=25. Cells are means; $^+/^-$: paired difference from E_2 on the same trials significantly better/worse (95% CI over observation sequences excludes 0). Bold marks the best trigger for that method and N. A dash means not run (dense methods stop at N=128).

Table 13: Stochastic volatility: predictive NLL (lower is better), common threshold $0.5N$. Resample when $E_2 = 1/\sum_i w_i^2$, $E_H = \exp(-\sum_i w_i \log w_i)$ or $E_\infty = 1/\max_i w_i$ falls below $0.5N$. $E_\infty \le E_2 \le E_H$, so E_∞ resamples most often and E_H least. $CV^2 > 1$ is the same rule as $E_2 < 0.5N$. 50 steps; 100 filter runs over 10 observation sequences (10 for OT and Gumbel).

	E_2				E_H				E_∞			
Method	32	128	512	2048	32	128	512	2048	32	128	512	2048
DSR (τ=0.1, paper)	1.66	1.66	1.66	1.66	1.66	1.66	1.66	1.66	**1.66**	**1.66**	**1.66**	**1.66**
DSR (τ=0.1, unsorted)	**1.67**	1.66	**1.66**	**1.66**	1.67	**1.66**	1.66	1.66	1.67	1.67	1.66	1.66
DSR (τ=0.5)	**1.67**	1.66	**1.66**	**1.66**	1.67	**1.66**	1.66	1.66	1.67	1.66	1.66	1.66
DSR (τ=2.5/N)†	1.66	1.66	**1.66**	1.66	1.66	1.66	1.66	1.66	**1.66**	**1.66**$^+$	1.66	**1.66**
Systematic	1.66	1.66	**1.66**	1.66	1.66	**1.66**	1.66	1.66	**1.66**	1.66	1.66	**1.66**
Multinomial	1.66	1.66	1.66	**1.66**	1.66	**1.66**	**1.66**	1.66	**1.66**	1.66	1.66	1.66
Systematic, stop-gradient (PyDPF)	1.66	1.66	1.66	1.66	1.66	**1.66**	**1.66**	1.66	**1.66**	1.66	1.66	**1.66**
Soft (α=0.5)	1.66	1.66	1.66	**1.66**	**1.66**	1.66	1.66	1.66	1.66	**1.66**	**1.66**	1.66
OT (PyDPF)	**1.66**	1.66	–	–	1.66	1.66	–	–	1.66	**1.66**	–	–
OT (PFJAX)	**1.66**	1.66	–	–	1.66	1.66	–	–	1.66	**1.66**	–	–
OT (FilterFlow)	1.66	1.66	–	–	**1.66**	1.66	–	–	1.66	**1.66**	–	–
Eigen moment matching	1.66	1.66	1.66	**1.66**	1.66	1.66	1.66	1.66	**1.66**	**1.66**	**1.66**	**1.66**
Gaussian moment matching	**1.66**	1.66	1.66	1.66	1.66	1.66	1.66	1.66	1.66	**1.66**	**1.66**	**1.66**
Gumbel-softmax	**1.66**	1.66	–	–	1.67	1.66	–	–	1.67	**1.66**	–	–
Variational (KL)	1.66	1.66	1.66	1.66	1.66$^-$	1.66	1.66	1.66	**1.66**	**1.65**	**1.65**	**1.65**

†Local diagnostic, not from the paper: τ rescaled so that τN equals the paper's value at N=25. Cells are means; $^+/^-$: paired difference from E_2 on the same trials significantly better/worse (95% CI over observation sequences excludes 0). Bold marks the best trigger for that method and N. A dash means not run (dense methods stop at N=128).

Table 14: Stochastic volatility: resampling steps out of 50, common threshold $0.5N$. Resample when $E_2 = 1/\sum_i w_i^2$, $E_H = \exp(-\sum_i w_i \log w_i)$ or $E_\infty = 1/\max_i w_i$ falls below $0.5N$. $E_\infty \le E_2 \le E_H$, so E_∞ resamples most often and E_H least. $CV^2 > 1$ is the same rule as $E_2 < 0.5N$. 50 steps; 100 filter runs over 10 observation sequences (10 for OT and Gumbel).

	E_2				E_H				E_∞			
Method	32	128	512	2048	32	128	512	2048	32	128	512	2048
DSR (τ=0.1, paper)	3.89	4.16	4.27	4.27	2.82	2.91	2.84	2.88	9.69	11.1	12.3	13.3
DSR (τ=0.1, unsorted)	3.7	3.79	3.91	3.93	2.7	2.67	2.68	2.7	8.78	10.1	11.1	12
DSR (τ=0.5)	3.63	3.81	3.92	3.94	2.66	2.66	2.68	2.7	8.75	10.1	11.1	12.1
DSR (τ=2.5/N)†	3.91	4.22	4.27	4.26	2.84	2.96	2.94	3.01	9.82	12.2	13.5	15.2
Systematic	4	4.25	4.24	4.31	2.84	2.96	2.96	3.01	10.3	12.3	13.8	15.5
Multinomial	3.9	4.27	4.35	4.27	2.97	3	2.99	2.99	10.2	12.4	13.8	15.4
Systematic, stop-gradient (PyDPF)	4.03	4.28	4.29	4.27	2.88	2.99	2.99	3.01	10.4	12.2	13.9	15.5
Soft (α=0.5)	5.63	5.71	5.8	5.87	3.88	4.1	4.22	4.31	14.3	17	19.4	21
OT (PyDPF)	4.1	4.2	–	–	3.1	3.2	–	–	10.3	12.3	–	–
OT (PFJAX)	4.1	4.3	–	–	3.1	3.2	–	–	10.1	12.3	–	–
OT (FilterFlow)	4.1	4.3	–	–	3.1	3.2	–	–	10.3	11.9	–	–
Eigen moment matching	4.02	4.26	4.29	4.28	2.93	2.97	3	2.99	10.6	12.5	14.1	15.7
Gaussian moment matching	3.97	4.15	4.31	4.24	2.92	2.93	2.98	2.97	10.6	12.5	14.2	15.8
Gumbel-softmax	3.5	4.4	–	–	2.9	2.9	–	–	9.5	10.9	–	–
Variational (KL)	3.97	4.24	4.25	4.27	2.84	2.97	2.97	2.97	10.4	12.3	14.1	15.8

†Local diagnostic, not from the paper: τ rescaled so that τN equals the paper's value at N=25. Cells are means; $^+/^-$: paired difference from E_2 on the same trials significantly better/worse (95% CI over observation sequences excludes 0). Bold marks the best trigger for that method and N. A dash means not run (dense methods stop at N=128).

Table 15: Bearing-only tracking: state RMSE (lower is better), common threshold $0.5N$. Resample when $E_2 = 1/\sum_i w_i^2$, $E_H = \exp(-\sum_i w_i \log w_i)$ or $E_\infty = 1/\max_i w_i$ falls below $0.5N$. $E_\infty \le E_2 \le E_H$, so E_∞ resamples most often and E_H least. $CV^2 > 1$ is the same rule as $E_2 < 0.5N$. 50 steps; 100 filter runs over 10 observation sequences (10 for OT and Gumbel).

	E_2				E_H				E_∞			
Method	32	128	512	2048	32	128	512	2048	32	128	512	2048
DSR (τ=0.1, paper)	**5.12**	4.83	4.87	5.19	5.21	**4.8**	**4.67**$^+$	**4.78**	5.38	5.01	5.23$^-$	5.41
DSR (τ=0.1, unsorted)	4.83	4.95	4.84	4.91	**4.78**	**4.89**	**4.73**$^+$	**4.88**	5.06	5.1	5.09	5.23
DSR (τ=0.5)	5.14	4.81	4.87	4.9	**5.12**	**4.75**	**4.78**	**4.86**	5.38$^-$	5.09	5.13	5.22
DSR (τ=2.5/N)†	**5.11**	**5.04**	4.81	4.42	5.2	5.2	**4.7**	4.45	5.48	5.46	4.79	**4.29**
Systematic	**5.08**	**4.55**	**4.13**	4.14	5.34	4.65	4.18	**4.13**	5.35	4.55	4.47	4.22
Multinomial	5.38	4.79	4.21	4.16	**5.33**	**4.73**	4.3	4.25	5.47	4.96	**4.2**	**4.15**
Systematic, stop-gradient (PyDPF)	5.5	4.76	**4.11**	4.11	5.41	**4.73**	4.26$^-$	4.15	**5.27**	4.84	4.2	**4.11**
Soft (α=0.5)	5.62	**4.8**	4.28	4.2	**5.36**	5.06	4.24	**4.18**	5.61	4.9	**4.21**	4.25
OT (PyDPF)	5.31	5.65	–	–	5.43	5.69	–	–	**5.11**	**5.37**	–	–
OT (PFJAX)	**4.97**	5.56	–	–	5.19	5.73	–	–	5.42	**5.4**	–	–
OT (FilterFlow)	5.22	**5.3**	–	–	**5.16**	5.75	–	–	5.23	5.71	–	–
Eigen moment matching	5.12	4.51	**4.21**	**4.24**	5.14	**4.48**	4.42	4.5	**5.05**	4.59	4.46	4.48
Gaussian moment matching	**5.23**	4.51	**4.36**	**4.2**	5.39	**4.48**	4.38	4.38	5.27	4.75	4.7	4.45
Gumbel-softmax	**4.45**	5.19	–	–	4.7	5.26	–	–	5.26	**5.13**	–	–
Variational (KL)	**5.19**	4.66	4.2	**4.27**	5.37	**4.62**	**4.18**	4.33	5.4	4.85	4.36	4.33

†Local diagnostic, not from the paper: τ rescaled so that τN equals the paper's value at N=25. Cells are means; $^+/^-$: paired difference from E_2 on the same trials significantly better/worse (95% CI over observation sequences excludes 0). Bold marks the best trigger for that method and N. A dash means not run (dense methods stop at N=128).

Table 16: Bearing-only tracking: RMSE after the observation gap (lower is better), common threshold $0.5N$. Resample when $E_2 = 1/\sum_i w_i^2$, $E_H = \exp(-\sum_i w_i \log w_i)$ or $E_\infty = 1/\max_i w_i$ falls below $0.5N$. $E_\infty \le E_2 \le E_H$, so E_∞ resamples most often and E_H least. $CV^2 > 1$ is the same rule as $E_2 < 0.5N$. 50 steps; 100 filter runs over 10 observation sequences (10 for OT and Gumbel).

	E_2				E_H				E_∞			
Method	32	128	512	2048	32	128	512	2048	32	128	512	2048
DSR (τ=0.1, paper)	**9.69**	9.25	9.31	9.97	9.92	**9.19**	**8.87**	**9.1**	10.2	9.55	10	10.4
DSR (τ=0.1, unsorted)	9.15	9.52	9.33	9.46	**9.03**	**9.4**	**9.12**	**9.43**	9.59	9.72	9.67	9.92
DSR (τ=0.5)	9.77	9.22	9.38	9.44	**9.76**	**9.13**	**9.2**	**9.37**	10.2	9.63	9.71	9.88
DSR (τ=2.5/N)†	**9.66**	**9.53**	9.03	8.14	9.88	9.86	**8.75**	8.22	10.4	10.6	8.91	**7.79**
Systematic	**9.54**	**8.44**	**7.61**	7.58	10.1	8.66	7.65	**7.55**	10.1	8.51	8.36	7.81
Multinomial	10.1	8.92	**7.76**	7.65	**9.99**	**8.81**	7.96	7.87	10.4	9.24	7.81	**7.6**
Systematic, stop-gradient (PyDPF)	10.4	8.96	**7.57**	7.54	10.2	**8.81**	7.96	7.65	**9.88**	9.13	7.79	**7.53**
Soft (α=0.5)	10.6	**9.01**	7.92	7.71	**10.1**	9.49	7.78	**7.67**	10.6	9.31	**7.74**	7.73
OT (PyDPF)	9.98	10.9	–	–	10.1	11.1	–	–	**9.59**	**10.3**	–	–
OT (PFJAX)	**9.19**	10.7	–	–	9.6	11.2	–	–	10.2	**10.3**	–	–
OT (FilterFlow)	9.96	**10.2**	–	–	9.99	11.3	–	–	**9.94**	11.2	–	–
Eigen moment matching	9.59	**8.28**	**7.59**	**7.69**	9.65	8.3	8.14	8.32	**9.51**	8.55	8.15	8.22
Gaussian moment matching	**9.74**	8.3	**7.98**	**7.62**	10.1	**8.27**	8.06	8.1	9.93	8.93	8.7	8.19
Gumbel-softmax	**8.26**	9.89	–	–	9.06	10	–	–	10.1	**9.88**	–	–
Variational (KL)	**9.81**	8.76	7.73	**7.9**	10.2	**8.63**	**7.68**	8.07	10.1	9.13	8.12	8.09

†Local diagnostic, not from the paper: τ rescaled so that τN equals the paper's value at N=25. Cells are means; $^+/^-$: paired difference from E_2 on the same trials significantly better/worse (95% CI over observation sequences excludes 0). Bold marks the best trigger for that method and N. A dash means not run (dense methods stop at N=128).

Table 17: Bearing-only tracking: 95% coverage (closer to 0.95 is better), common threshold $0.5N$. Resample when $E_2 = 1/\sum_i w_i^2$, $E_H = \exp(-\sum_i w_i \log w_i)$ or $E_\infty = 1/\max_i w_i$ falls below $0.5N$. $E_\infty \le E_2 \le E_H$, so E_∞ resamples most often and E_H least. $CV^2 > 1$ is the same rule as $E_2 < 0.5N$. 50 steps; 100 filter runs over 10 observation sequences (10 for OT and Gumbel).

	E_2				E_H				E_∞			
Method	32	128	512	2048	32	128	512	2048	32	128	512	2048
DSR (τ=0.1, paper)	0.0692	0.0836	0.0874	0.0572	**0.0724**	**0.108**	**0.113**	**0.0794**	0.0326$^-$	0.0272$^-$	0.0216$^-$	0.0178
DSR (τ=0.1, unsorted)	0.0422	0.0202	0.0204	0.0174	**0.0574**$^+$	**0.0274**	**0.0284**	**0.0236**	0.0222$^-$	0.017$^-$	0.0168	0.0162
DSR (τ=0.5)	0.02	0.0234	0.0216	0.02	**0.0302**	**0.0278**	**0.0274**	**0.0256**	0.0146	0.0168	0.0172	0.016
DSR (τ=2.5/N)†	0.0762	0.299	0.494	0.682	**0.0808**	**0.301**	**0.509**	**0.709**	0.0408$^-$	0.205$^-$	0.413$^-$	0.565$^-$
Systematic	0.265	0.636	0.826	**0.904**	0.253	**0.654**	**0.838**	0.901	**0.269**	0.628	0.789	0.895
Multinomial	0.196	0.54	**0.809**	**0.901**	**0.218**	**0.551**	0.799	0.89	0.181	0.509	0.791	0.872
Systematic, stop-gradient (PyDPF)	0.21	**0.649**	**0.85**	**0.904**	0.239	0.639	0.825	0.897	**0.243**$^+$	0.614	0.825	0.899
Soft (α=0.5)	**0.179**	0.519	**0.802**	0.891	0.174	**0.537**	0.778$^-$	**0.894**	0.148	0.496	0.789	0.862
OT (PyDPF)	0.09	0.04	–	–	**0.1**	**0.102**$^+$	–	–	0.032	0.024	–	–
OT (PFJAX)	0.092	0.042	–	–	**0.116**	**0.1**	–	–	0.044	0.024$^-$	–	–
OT (FilterFlow)	0.054	0.052	–	–	**0.066**	**0.07**	–	–	0.016	0.022	–	–
Eigen moment matching	0.51	0.778	0.87	**0.901**	0.481	**0.788**	**0.876**	0.882	**0.513**	0.755	0.861	0.859
Gaussian moment matching	0.362	0.724	0.85	**0.902**	**0.366**	**0.752**	**0.858**	0.892	0.344	0.677	0.811$^-$	0.864
Gumbel-softmax	0.196	0.216	–	–	**0.252**	**0.26**	–	–	0.084	0.142$^-$	–	–
Variational (KL)	0.246	0.633	**0.83**	**0.873**	**0.255**	**0.649**	**0.83**	0.864	0.237	0.577	0.755$^-$	0.837

†Local diagnostic, not from the paper: τ rescaled so that τN equals the paper's value at N=25. Cells are means; $^+/^-$: paired difference from E_2 on the same trials significantly better/worse (95% CI over observation sequences excludes 0). Bold marks the best trigger for that method and N. A dash means not run (dense methods stop at N=128).

Table 18: Bearing-only tracking: predictive NLL (lower is better), common threshold $0.5N$. Resample when $E_2 = 1/\sum_i w_i^2$, $E_H = \exp(-\sum_i w_i \log w_i)$ or $E_\infty = 1/\max_i w_i$ falls below $0.5N$. $E_\infty \le E_2 \le E_H$, so E_∞ resamples most often and E_H least. $CV^2 > 1$ is the same rule as $E_2 < 0.5N$. 50 steps; 100 filter runs over 10 observation sequences (10 for OT and Gumbel).

	E_2				E_H				E_∞			
Method	32	128	512	2048	32	128	512	2048	32	128	512	2048
DSR (τ=0.1, paper)	−0.528	−0.638	−0.612	−0.435	**−0.603**	**−0.788**	**−0.736**	**−0.753**	0.302$^-$	−0.153$^-$	0.0226$^-$	0.309$^-$
DSR (τ=0.1, unsorted)	−0.385	−0.536	−0.565	−0.646	**−0.571**$^+$	**−0.722**$^+$	**−0.71**$^+$	**−0.735**$^+$	0.38$^-$	0.136$^-$	0.4$^-$	0.714$^-$
DSR (τ=0.5)	0.104	−0.454	−0.545	−0.611	**−0.194**$^+$	**−0.589**$^+$	**−0.736**	**−0.682**	1.26$^-$	0.569$^-$	0.526$^-$	0.742$^-$
DSR (τ=2.5/N)†	−0.613	**−1.16**	**−1.27**	**−1.31**	**−0.701**	−1.15	−1.24	−1.3	0.0257$^-$	−0.948	−1.18	−1.25
Systematic	−1.2	−1.31	−1.32	−1.33	−1.21	**−1.31**	**−1.32**	**−1.33**	**−1.21**	−1.3	−1.32	−1.33
Multinomial	−1.09	−1.3	−1.32	**−1.33**	**−1.1**	**−1.3**	**−1.33**	−1.32	−1.08	−1.29	−1.32	−1.32
Systematic, stop-gradient (PyDPF)	−1.2	−1.31	**−1.32**	**−1.33**	−1.14	**−1.31**	−1.32	−1.33	**−1.22**	−1.31	−1.32	−1.33
Soft (α=0.5)	**−1**	−1.29	−1.32	**−1.33**	−0.94	−1.29	**−1.32**	−1.33	0.145	**−1.3**	−1.32	−1.33$^-$
OT (PyDPF)	−0.865	**−1.14**	–	–	**−0.989**	−0.896	–	–	−0.837	−0.849	–	–
OT (PFJAX)	−0.905	**−1.14**	–	–	**−0.995**	−0.89	–	–	−0.743	−0.878	–	–
OT (FilterFlow)	**−1.08**	−0.797	–	–	−0.934	**−1.06**	–	–	−0.688$^-$	−0.648	–	–
Eigen moment matching	**−1.27**	−1.32	−1.32	−1.32	−1.26	−1.32	−1.32	−1.32	1.06	**−1.32**	**−1.33**	**−1.33**
Gaussian moment matching	−1.27	−1.32	−1.32	−1.33	−1.26	**−1.32**	**−1.33**	−1.33$^+$	**−1.27**	−1.31	−1.32	**−1.33**
Gumbel-softmax	**−1.21**	−1.23	–	–	−1.02	**−1.24**	–	–	−0.951	−1.22	–	–
Variational (KL)	**−1.2**	**−1.31**	−1.32	**−1.32**	−1.19	−1.3	**−1.32**	−1.32	−1.16	−1.25	−1.31	−1.31

†Local diagnostic, not from the paper: τ rescaled so that τN equals the paper's value at N=25. Cells are means; $^+/^-$: paired difference from E_2 on the same trials significantly better/worse (95% CI over observation sequences excludes 0). Bold marks the best trigger for that method and N. A dash means not run (dense methods stop at N=128).

Table 19: Bearing-only tracking: resampling steps out of 50, common threshold $0.5N$. Resample when $E_2 = 1/\sum_i w_i^2$, $E_H = \exp(-\sum_i w_i \log w_i)$ or $E_\infty = 1/\max_i w_i$ falls below $0.5N$. $E_\infty \le E_2 \le E_H$, so E_∞ resamples most often and E_H least. $CV^2 > 1$ is the same rule as $E_2 < 0.5N$. 50 steps; 100 filter runs over 10 observation sequences (10 for OT and Gumbel).

	E_2				E_H				E_∞			
Method	32	128	512	2048	32	128	512	2048	32	128	512	2048
DSR (τ=0.1, paper)	7.21	7.3	7.52	7.88	6.22	6.15	6.18	6.41	11.7	12.9	13.9	15.3
DSR (τ=0.1, unsorted)	7.19	6.81	6.57	6.39	6.03	5.69	5.73	5.51	11.3	12.1	12.8	13.4
DSR (τ=0.5)	6.64	6.4	6.34	6.39	5.52	5.55	5.47	5.46	10.2	11	11.9	12.9
DSR (τ=2.5/N)†	7.5	8.03	8.29	8.67	6.39	6.98	7.24	7.11	12	13.9	15	15.4
Systematic	8.85	9.35	9.48	9.35	7.54	7.72	7.79	7.92	15	16.3	17	17.1
Multinomial	9.03	9.39	9.44	9.5	7.55	7.81	7.72	7.88	14.4	16	16.9	17.1
Systematic, stop-gradient (PyDPF)	9	9.41	9.42	9.43	7.51	7.79	7.82	7.94	14.8	16.3	16.9	17.1
Soft (α=0.5)	13.5	14.1	14.3	14.4	11.8	12.2	12.3	12.3	20.3	22.4	23.3	23.9
OT (PyDPF)	8.4	7.9	–	–	6.9	6.5	–	–	12.5	14.2	–	–
OT (PFJAX)	8.3	7.9	–	–	6.9	6.6	–	–	13.2	13.8	–	–
OT (FilterFlow)	6.9	8.1	–	–	6.5	6.6	–	–	12.6	13.8	–	–
Eigen moment matching	9.48	9.51	9.57	9.45	7.87	8.01	7.92	8.07	16.4	16.8	17.2	17.6
Gaussian moment matching	9.72	9.61	9.55	9.51	7.95	8.01	7.92	7.98	15.7	16.7	17.1	17.6
Gumbel-softmax	7.7	8.4	–	–	6.7	7	–	–	13.8	13.8	–	–
Variational (KL)	8.8	9.28	9.33	9.35	7.38	7.8	7.77	7.89	14.8	16	16.4	16.8

†Local diagnostic, not from the paper: τ rescaled so that τN equals the paper's value at N=25. Cells are means; $^+/^-$: paired difference from E_2 on the same trials significantly better/worse (95% CI over observation sequences excludes 0). Bold marks the best trigger for that method and N. A dash means not run (dense methods stop at N=128).

Table 20: Stochastic Lorenz-63: state RMSE (lower is better), common threshold $0.5N$. Resample when $E_2 = 1/\sum_i w_i^2$, $E_H = \exp(-\sum_i w_i \log w_i)$ or $E_\infty = 1/\max_i w_i$ falls below $0.5N$. $E_\infty \le E_2 \le E_H$, so E_∞ resamples most often and E_H least. $CV^2 > 1$ is the same rule as $E_2 < 0.5N$. 50 steps; 100 filter runs over 10 observation sequences (10 for OT and Gumbel).

	E_2				E_H				E_∞			
Method	32	128	512	2048	32	128	512	2048	32	128	512	2048
DSR (τ=0.1, paper)	0.708	0.542	0.491	0.486	**0.606**	**0.514**	**0.481**	**0.476**	0.831	0.704	0.584	0.521
DSR (τ=0.1, unsorted)	0.853	0.619	0.538	0.514	**0.69**	**0.551**	**0.506**	**0.495**	2.21	1.87	1.93	1.06
DSR (τ=0.5)	1.22	0.792	0.548	0.52	**0.91**	**0.573**	**0.517**	**0.498**	3.22	2.04	1.7	0.973
DSR (τ=2.5/N)†	0.64	0.479	0.446	0.433	**0.6**	**0.477**	**0.442**	**0.432**	0.748	0.53	0.45	0.435
Systematic	0.563	**0.456**	0.437	0.431	0.558	0.46	0.439	0.431	**0.551**	0.457	**0.436**	**0.43**
Multinomial	**0.549**	0.469	0.44	0.43	0.577	0.465	0.438	0.432^-	0.585	**0.462**	**0.436**	**0.43**
Systematic, stop-gradient (PyDPF)	0.584	0.464	0.439	**0.431**	0.572	0.466	**0.436**	0.431	**0.559**	$\mathbf{0.456}^+$	0.438	0.431
Soft (α=0.5)	**0.616**	**0.467**	0.44	**0.43**	0.644	0.469	0.441	0.432^-	0.634	0.473	**0.439**	0.431
OT (PyDPF)	**0.542**	0.477	–	–	0.786	**0.471**	–	–	0.574	0.479	–	–
OT (PFJAX)	0.55	0.468	–	–	0.785	**0.455**	–	–	**0.536**	0.469	–	–
OT (FilterFlow)	0.595	0.491	–	–	0.804	**0.467**	–	–	**0.585**	0.499	–	–
Eigen moment matching	0.533	**0.456**	**0.437**	**0.43**	0.533	0.457	0.439	0.432	**0.529**	0.46	0.44	0.432
Gaussian moment matching	0.591	**0.46**	0.439	**0.43**	0.592	0.467^-	0.44	0.432	**0.589**	0.46	**0.439**	0.432
Gumbel-softmax	0.756	0.484	–	–	**0.547**	**0.474**	–	–	0.777	0.497	–	–
Variational (KL)	0.546	0.461	0.438	**0.429**	**0.542**	0.466	**0.438**	0.431	0.571	**0.46**	0.439	0.432

†Local diagnostic, not from the paper: τ rescaled so that τN equals the paper's value at N=25. Cells are means; $^+/^-$: paired difference from E_2 on the same trials significantly better/worse (95% CI over observation sequences excludes 0). Bold marks the best trigger for that method and N. A dash means not run (dense methods stop at N=128).

Table 21: Stochastic Lorenz-63: 95% coverage (closer to 0.95 is better), common threshold $0.5N$. Resample when $E_2 = 1/\sum_i w_i^2$, $E_H = \exp(-\sum_i w_i \log w_i)$ or $E_\infty = 1/\max_i w_i$ falls below $0.5N$. $E_\infty \le E_2 \le E_H$, so E_∞ resamples most often and E_H least. $CV^2 > 1$ is the same rule as $E_2 < 0.5N$. 50 steps; 100 filter runs over 10 observation sequences (10 for OT and Gumbel).

	E_2				E_H				E_∞			
Method	32	128	512	2048	32	128	512	2048	32	128	512	2048
DSR (τ=0.1, paper)	0.397	0.579	0.655	0.682	$\mathbf{0.443}^+$	$\mathbf{0.613}^+$	$\mathbf{0.688}^+$	$\mathbf{0.717}^+$	0.36^-	0.515^-	0.568^-	0.606^-
DSR (τ=0.1, unsorted)	0.378	0.516	0.576	0.597	**0.412**	**0.555**	**0.615**	**0.644**	0.308^-	0.428^-	0.462^-	0.488^-
DSR (τ=0.5)	0.332	0.489	0.563	0.593	**0.36**	**0.534**	**0.599**	**0.636**	0.266^-	0.399^-	0.453^-	0.488^-
DSR (τ=2.5/N)†	0.418	0.723	0.877	0.932	**0.431**	$\mathbf{0.741}^+$	**0.887**	**0.934**	0.385^-	0.674^-	0.849^-	0.922
Systematic	0.58	0.854	0.933	**0.95**	0.57	0.845	0.924	0.951	$\mathbf{0.601}^+$	**0.854**	$\mathbf{0.934}^+$	0.948
Multinomial	**0.581**	0.833	0.928	0.95	0.564	**0.838**	**0.932**	0.948	0.571	**0.838**	0.93	**0.95**
Systematic, stop-gradient (PyDPF)	0.568	0.849	0.928	**0.95**	0.577	0.844	**0.93**	0.949	$\mathbf{0.595}^+$	$\mathbf{0.861}^+$	0.924	0.949
Soft (α=0.5)	**0.473**	0.814	0.922	**0.95**	0.471	**0.817**	**0.925**	0.949	0.472	0.811	0.923	0.951
OT (PyDPF)	0.506	0.772	–	–	0.448	**0.778**	–	–	**0.524**	0.77	–	–
OT (PFJAX)	0.532	0.804	–	–	0.48	0.808	–	–	**0.548**	**0.82**	–	–
OT (FilterFlow)	0.484	0.734	–	–	0.434	**0.766**	–	–	**0.496**	0.704	–	–
Eigen moment matching	0.612	**0.858**	**0.932**	0.949	0.599	0.852	0.929	**0.951**	$\mathbf{0.622}^+$	**0.858**	0.93	0.949
Gaussian moment matching	**0.568**	0.847	**0.929**	0.947	0.561	0.836^-	0.929	0.946	0.566	**0.849**	0.928	**0.948**
Gumbel-softmax	0.464	0.702	–	–	**0.526**	**0.734**	–	–	0.444	0.672	–	–
Variational (KL)	0.588	0.845	**0.927**	**0.949**	0.585	0.842	0.924	0.949	**0.593**	**0.852**	0.926	0.946

†Local diagnostic, not from the paper: τ rescaled so that τN equals the paper's value at N=25. Cells are means; $^+/^-$: paired difference from E_2 on the same trials significantly better/worse (95% CI over observation sequences excludes 0). Bold marks the best trigger for that method and N. A dash means not run (dense methods stop at N=128).

Table 22: Stochastic Lorenz-63: predictive NLL (lower is better), common threshold $0.5N$. Resample when $E_2 = 1/\sum_i w_i^2$, $E_H = \exp(-\sum_i w_i \log w_i)$ or $E_\infty = 1/\max_i w_i$ falls below $0.5N$. $E_\infty \le E_2 \le E_H$, so E_∞ resamples most often and E_H least. $CV^2 > 1$ is the same rule as $E_2 < 0.5N$. 50 steps; 100 filter runs over 10 observation sequences (10 for OT and Gumbel).

	E_2				E_H				E_∞			
Method	32	128	512	2048	32	128	512	2048	32	128	512	2048
DSR (τ=0.1, paper)	35.3	31.8	31.4	31.4	**32.8**	**31.5**	**31.3**	**31.3**	37.5	35.3	32.4	31.6
DSR (τ=0.1, unsorted)	38.4	33.4	31.8	31.6	**34.5**	**31.9**	**31.5**	**31.4**	65.7	59.7	57.9	43.6
DSR (τ=0.5)	46	36.6	31.9	31.6	**37.7**	**32.2**	**31.6**	**31.4**	92.3	63.1	52.7	41.4
DSR (τ=2.5/N)†	33.7	31.3	31.2	31.2	**32.4**	**31.3**	**31.2**$^+$	**31.2**$^+$	35.9	31.9	31.2	31.2
Systematic	31.8	31.2	**31.2**	31.2	31.7	31.2	31.2	**31.2**	**31.7**	**31.2**	31.2	31.2
Multinomial	**31.6**	31.2	31.2	**31.2**	32	31.2	31.2	31.2	32.2	**31.2**	**31.2**	31.2
Systematic, stop-gradient (PyDPF)	32.2	31.2	31.2	**31.2**	32	31.2	**31.2**	31.2	**31.8**	**31.2**	31.2	31.2$^-$
Soft (α=0.5)	**32.2**	**31.2**	**31.2**	**31.2**	32.7	31.2	31.2	**31.2**	32.5	31.3	**31.2**	31.2
OT (PyDPF)	**31.5**	31.3	–	–	35.6	**31.3**	–	–	31.6	31.3	–	–
OT (PFJAX)	31.5	31.3	–	–	36.1	**31.3**	–	–	**31.5**	31.3	–	–
OT (FilterFlow)	**31.7**	31.4	–	–	35.6	**31.3**	–	–	31.7	31.4	–	–
Eigen moment matching	31.5	**31.2**	**31.2**	**31.2**	31.6	31.2	31.2	31.2$^-$	**31.5**	31.2	31.2	31.2
Gaussian moment matching	32.3	31.2	31.2	**31.2**	32.3	31.2$^-$	31.2	31.2$^-$	**32.2**	**31.2**	**31.2**	**31.2**
Gumbel-softmax	35.4	31.3	–	–	**31.6**	**31.3**	–	–	36	31.4	–	–
Variational (KL)	**31.6**	31.2	31.2	**31.2**	31.7	31.2	31.2	31.2	32	**31.2**	**31.2**	31.2

†Local diagnostic, not from the paper: τ rescaled so that τN equals the paper's value at N=25. Cells are means; $^+/^-$: paired difference from E_2 on the same trials significantly better/worse (95% CI over observation sequences excludes 0). Bold marks the best trigger for that method and N. A dash means not run (dense methods stop at N=128).

Table 23: Stochastic Lorenz-63: resampling steps out of 50, common threshold $0.5N$. Resample when $E_2 = 1/\sum_i w_i^2$, $E_H = \exp(-\sum_i w_i \log w_i)$ or $E_\infty = 1/\max_i w_i$ falls below $0.5N$. $E_\infty \le E_2 \le E_H$, so E_∞ resamples most often and E_H least. $CV^2 > 1$ is the same rule as $E_2 < 0.5N$. 50 steps; 100 filter runs over 10 observation sequences (10 for OT and Gumbel).

	E_2				E_H				E_∞			
Method	32	128	512	2048	32	128	512	2048	32	128	512	2048
DSR (τ=0.1, paper)	38.9	39	38.9	38.7	33.9	33.9	33.6	33.5	48.2	48.6	48.9	49
DSR (τ=0.1, unsorted)	38.8	38.4	38.1	38	33.9	33.4	33.1	33	48.2	48.6	49.1	49.3
DSR (τ=0.5)	38.8	38.6	38.1	38.1	34	33.5	33.1	33	48.2	48.5	49	49.2
DSR (τ=2.5/N)†	39	40.3	41.4	42.4	34.2	34.9	35.8	36.3	48.3	49.1	49.5	49.6
Systematic	41.6	42.9	43.3	43.3	36.2	36.8	37	37.1	49.2	49.5	49.6	49.7
Multinomial	41.5	42.7	43.3	43.2	35.7	36.8	37.2	37	49.3	49.5	49.6	49.7
Systematic, stop-gradient (PyDPF)	41.4	43	43.1	43.3	36.3	36.8	37.2	37.1	49.2	49.5	49.6	49.7
Soft (α=0.5)	48.4	49.1	49.1	49	46.2	47.6	48.1	48	49.9	50	50	50
OT (PyDPF)	41.3	41	–	–	36	35.4	–	–	49	49.2	–	–
OT (PFJAX)	40.1	41.2	–	–	35.4	35.2	–	–	48.8	49.2	–	–
OT (FilterFlow)	40	40.7	–	–	36.1	35.2	–	–	48.9	49	–	–
Eigen moment matching	41.4	42.9	43.2	43.2	35.9	36.9	37.1	37.2	49.3	49.5	49.6	49.6
Gaussian moment matching	41.3	42.7	43.3	43.2	36.1	37	37.2	37.3	49.1	49.5	49.5	49.6
Gumbel-softmax	39.9	40.1	–	–	34	34.8	–	–	48.5	48.7	–	–
Variational (KL)	41.3	42.7	43.2	43.1	36	36.7	37	36.9	49.1	49.5	49.6	49.7

†Local diagnostic, not from the paper: τ rescaled so that τN equals the paper's value at N=25. Cells are means; $^+/^-$: paired difference from E_2 on the same trials significantly better/worse (95% CI over observation sequences excludes 0). Bold marks the best trigger for that method and N. A dash means not run (dense methods stop at N=128).

Table 24: Stochastic Lorenz-96: state RMSE (lower is better), common threshold $0.5N$. Resample when $E_2 = 1/\sum_i w_i^2$, $E_H = \exp(-\sum_i w_i \log w_i)$ or $E_\infty = 1/\max_i w_i$ falls below $0.5N$. $E_\infty \le E_2 \le E_H$, so E_∞ resamples most often and E_H least. $CV^2 > 1$ is the same rule as $E_2 < 0.5N$. 50 steps; 100 filter runs over 10 observation sequences (10 for OT and Gumbel).

	E_2				E_H				E_∞			
Method	32	128	512	2048	32	128	512	2048	32	128	512	2048
DSR (τ=0.1, paper)	3.09	2.46	2.11	1.9	**3.08**	**2.43**$^+$	**2.08**$^+$	**1.82**$^+$	3.09	2.47	2.12$^-$	1.91$^-$
DSR (τ=0.1, unsorted)	3.07	2.47	2.13	1.9	3.07	**2.44**$^+$	**2.08**$^+$	**1.82**$^+$	**3.07**	2.48	2.13$^-$	1.91$^-$
DSR (τ=0.5)	3.2	2.65	2.25	1.97	**3.19**	**2.62**$^+$	**2.2**$^+$	**1.91**$^+$	3.21	2.65	2.25$^-$	1.99$^-$
DSR (τ=2.5/N)†	3.02	2.17	1.45	1.08	**3**	**2.15**$^+$	**1.44**	**1.07**$^+$	3.01	2.17	1.45$^-$	1.08
Systematic	2.76	2.09	1.45	**1.07**	**2.74**	**2.09**	**1.45**	1.07	2.76	2.09	1.45	1.07
Multinomial	**2.8**	2.06	**1.41**	1.09	2.81	**2.06**	1.41	1.09	2.8	2.06	1.41	**1.09**
Systematic, stop-gradient (PyDPF)	2.73	**2.05**	1.41	**1.06**	2.73	2.05	1.41	1.06	**2.73**	2.05	**1.41**	1.06
Soft (α=0.5)	**2.96**	2.14	**1.48**	**1.07**	2.96	**2.14**	1.48	**1.07**	**2.96**	2.14	**1.48**	**1.07**
OT (PyDPF)	3.11	2.44	–	–	**3.08**	**2.42**	–	–	3.11	2.43	–	–
OT (PFJAX)	3.27	**2.28**	–	–	**3.2**	2.3	–	–	3.27	2.29	–	–
OT (FilterFlow)	3.36	2.52	–	–	**3.36**	**2.52**	–	–	3.36	2.52	–	–
Eigen moment matching	2.62	1.75	**1.1**	0.849	2.62	1.75	1.1	**0.849**	**2.61**	**1.75**	1.1	0.849
Gaussian moment matching	2.55	1.8	**1.15**	**0.865**	2.55	**1.79**	1.15	0.865	**2.55**	1.8	1.15	**0.865**
Gumbel-softmax	**3.15**	2.44	–	–	3.18	**2.41**	–	–	3.15	2.43	–	–
Variational (KL)	2.77	**2.05**	**1.41**	1.09	2.79$^-$	2.05	1.41	1.09	**2.77**	2.05	1.41	**1.09**

†Local diagnostic, not from the paper: τ rescaled so that τN equals the paper's value at N=25. Cells are means; $^+/^-$: paired difference from E_2 on the same trials significantly better/worse (95% CI over observation sequences excludes 0). Bold marks the best trigger for that method and N. A dash means not run (dense methods stop at N=128).

Table 25: Stochastic Lorenz-96: 95% coverage (closer to 0.95 is better), common threshold $0.5N$. Resample when $E_2 = 1/\sum_i w_i^2$, $E_H = \exp(-\sum_i w_i \log w_i)$ or $E_\infty = 1/\max_i w_i$ falls below $0.5N$. $E_\infty \le E_2 \le E_H$, so E_∞ resamples most often and E_H least. $CV^2 > 1$ is the same rule as $E_2 < 0.5N$. 50 steps; 100 filter runs over 10 observation sequences (10 for OT and Gumbel).

	E_2				E_H				E_∞			
Method	32	128	512	2048	32	128	512	2048	32	128	512	2048
DSR (τ=0.1, paper)	0	0.0024	0.004	0.0102	$\mathbf{2 \times 10^{-4}}$	**0.005**$^+$	**0.0168**$^+$	**0.0266**$^+$	0	8×10^{-4}	0.0016	0.003
DSR (τ=0.1, unsorted)	**0**	0.0016	0.0024	0.006	**0**	**0.0024**	**0.0136**$^+$	**0.0302**$^+$	**0**	4×10^{-4}	0.001	8×10^{-4}
DSR (τ=0.5)	**0**	4×10^{-4}	0.001	0.0054	**0**	**0.0018**	**0.0088**	**0.0202**$^+$	**0**	4×10^{-4}	4×10^{-4}	0.0012
DSR (τ=2.5/N)†	2×10^{-4}	0.008	0.0828	0.272	$\mathbf{4 \times 10^{-4}}$	**0.0182**$^+$	**0.118**$^+$	**0.331**$^+$	0	0.0058	0.0724	0.271
Systematic	0.0016	**0.0788**	**0.303**	**0.563**	**0.002**	0.0738	0.302	0.559	0.0012	0.0774	0.303	0.563
Multinomial	**0.001**	0.0634	**0.308**	**0.544**	8×10^{-4}	0.063	0.303	0.538	**0.001**	**0.064**	0.306	**0.544**
Systematic, stop-gradient (PyDPF)	**0.001**	**0.0852**	0.321	**0.56**	8×10^{-4}	0.072$^-$	0.31	0.55$^-$	**0.001**	0.0812	**0.323**	0.56
Soft (α=0.5)	**0.0012**	0.035	**0.249**	**0.519**	8×10^{-4}	**0.037**	0.248	**0.519**	**0.0012**	0.035	**0.249**	**0.519**
OT (PyDPF)	**0**	0.004	–	–	**0**	**0.016**	–	–	**0**	0.012	–	–
OT (PFJAX)	**0**	**0.034**	–	–	**0**	0.028	–	–	**0**	0.028	–	–
OT (FilterFlow)	**0**	0.002	–	–	**0**	0.002	–	–	**0**	**0.004**	–	–
Eigen moment matching	0.0052	**0.156**	**0.514**	0.713	0.0032	0.152	0.513	**0.714**	**0.0058**	0.155	0.514	**0.714**
Gaussian moment matching	0.0032	0.147	**0.48**	**0.706**	0.002	0.147	0.479	0.705	**0.0038**	**0.148**	0.48	**0.706**
Gumbel-softmax	**0.002**	0.016	–	–	**0.002**	**0.032**	–	–	**0.002**	0.012	–	–
Variational (KL)	**0.0022**	0.0704	0.323	0.543	0.0012	0.068	**0.324**	0.539	0.0016	**0.0728**	0.323	**0.544**

†Local diagnostic, not from the paper: τ rescaled so that τN equals the paper's value at N=25. Cells are means; $^+/^-$: paired difference from E_2 on the same trials significantly better/worse (95% CI over observation sequences excludes 0). Bold marks the best trigger for that method and N. A dash means not run (dense methods stop at N=128).

Table 26: Stochastic Lorenz-96: predictive NLL (lower is better), common threshold $0.5N$. Resample when $E_2 = 1/\sum_i w_i^2$, $E_H = \exp(-\sum_i w_i \log w_i)$ or $E_\infty = 1/\max_i w_i$ falls below $0.5N$. $E_\infty \le E_2 \le E_H$, so E_∞ resamples most often and E_H least. $CV^2 > 1$ is the same rule as $E_2 < 0.5N$. 50 steps; 100 filter runs over 10 observation sequences (10 for OT and Gumbel).

	E_2				E_H				E_∞			
Method	32	128	512	2048	32	128	512	2048	32	128	512	2048
DSR (τ=0.1, paper)	69.2	51.5	42.7	37.8	**69**	**51**$^+$	**42.2**$^+$	**36.8**$^+$	69.3	51.5	42.8$^-$	38$^-$
DSR (τ=0.1, unsorted)	68	52	43	38	68.1	**51.6**$^+$	**42.3**$^+$	**37**$^+$	**68**	52.1	43.1	38.2$^-$
DSR (τ=0.5)	72.1	56.5	45.8	39	**71.8**	**56.1**	**45.1**$^+$	**38.1**$^+$	72.2	56.4	45.9$^-$	39.2$^-$
DSR (τ=2.5/N)†	67.1	45	29.4	22.8	**66.7**	**44.8**$^+$	**29.3**$^+$	**22.8**$^+$	66.9	45	29.4	22.8
Systematic	60	**43.8**	30.1	**22.7**	**59.8**	43.8	**30.1**	22.7	60.1	43.8	30.1	22.7
Multinomial	**60.9**	43.1	**29.3**	**23.1**	61.2	**43**	29.3$^-$	23.2	60.9	43	29.3	**23.1**
Systematic, stop-gradient (PyDPF)	59.3	**42.9**	29	22.6	**59.3**	42.9	29	**22.6**	59.3	42.9	**29**	22.6
Soft (α=0.5)	**65.7**	44.9	**30.2**	**22.4**	65.7	**44.9**	30.2	**22.4**	**65.7**	44.9	**30.2**	**22.4**
OT (PyDPF)	68.1	54.7	–	–	**67.5**	**54.5**	–	–	68.1	54.6	–	–
OT (PFJAX)	73.2	**49.5**	–	–	**72.1**	49.7	–	–	73.2	49.6	–	–
OT (FilterFlow)	76.2	56.4	–	–	76.2	**56.4**	–	–	**76.2**	56.4	–	–
Eigen moment matching	57	36	**23.2**	19.4	57	36	23.2	**19.4**	**57**	**36**	23.2	19.4
Gaussian moment matching	54	37	**24.3**	**19.9**	54.3	37	24.3	19.9	**54**	**37**	24.3	**19.9**
Gumbel-softmax	**70.5**	55.1	–	–	71	**54.9**	–	–	70.5	55	–	–
Variational (KL)	60.2	42.7	29	23.2	60.5$^-$	42.8$^-$	29	23.2	**60.1**	**42.7**	**29**	**23.2**

†Local diagnostic, not from the paper: τ rescaled so that τN equals the paper's value at N=25. Cells are means; $^+/^-$: paired difference from E_2 on the same trials significantly better/worse (95% CI over observation sequences excludes 0). Bold marks the best trigger for that method and N. A dash means not run (dense methods stop at N=128).

Table 27: Stochastic Lorenz-96: resampling steps out of 50, common threshold $0.5N$. Resample when $E_2 = 1/\sum_i w_i^2$, $E_H = \exp(-\sum_i w_i \log w_i)$ or $E_\infty = 1/\max_i w_i$ falls below $0.5N$. $E_\infty \le E_2 \le E_H$, so E_∞ resamples most often and E_H least. $CV^2 > 1$ is the same rule as $E_2 < 0.5N$. 50 steps; 100 filter runs over 10 observation sequences (10 for OT and Gumbel).

	E_2				E_H				E_∞			
Method	32	128	512	2048	32	128	512	2048	32	128	512	2048
DSR (τ=0.1, paper)	49.3	49.1	49.1	48.8	47.4	45.8	45	43.7	50	50	50	50
DSR (τ=0.1, unsorted)	49.4	49.3	49.2	49	47.6	45.9	45	43.1	50	50	50	50
DSR (τ=0.5)	49.4	49.4	49.3	49	48	46.5	45.2	44	50	50	50	50
DSR (τ=2.5/N)†	49.1	49.2	49.1	49.3	47	45.6	44.8	44.7	50	50	50	50
Systematic	49.6	49.8	49.8	49.9	47.6	48.4	48.6	48.6	50	50	50	50
Multinomial	49.6	49.8	49.9	50	48.1	48.1	48.5	48.7	50	50	50	50
Systematic, stop-gradient (PyDPF)	49.6	49.8	49.9	49.9	47.7	48.2	48.5	48.6	50	50	50	50
Soft (α=0.5)	50	50	50	50	49.9	50	50	50	50	50	50	50
OT (PyDPF)	49.7	49.6	–	–	48	46.8	–	–	50	50	–	–
OT (PFJAX)	49.9	49.5	–	–	48.3	46.7	–	–	50	50	–	–
OT (FilterFlow)	49.6	49.5	–	–	48.8	47	–	–	50	50	–	–
Eigen moment matching	49.5	49.8	49.9	49.9	47.6	48.4	48.5	48.7	50	50	50	50
Gaussian moment matching	49.6	49.8	49.9	49.9	47.3	48.5	48.6	48.6	50	50	50	50
Gumbel-softmax	49.4	49.5	–	–	48.4	47	–	–	50	50	–	–
Variational (KL)	49.6	49.8	49.8	49.9	47.7	48.4	48.3	48.5	50	50	50	50

†Local diagnostic, not from the paper: τ rescaled so that τN equals the paper's value at N=25. Cells are means; $^+/^-$: paired difference from E_2 on the same trials significantly better/worse (95% CI over observation sequences excludes 0). Bold marks the best trigger for that method and N. A dash means not run (dense methods stop at N=128).

Resampling triggers, matched resampling rate

Table 28: Linear Gaussian SSM: state RMSE (lower is better), matched resampling rate. Resample when $E_H < \tau_H N$ or $E_\infty < \tau_\infty N$, with per-model thresholds calibrated so that systematic resampling resamples as often as under $E_2 < 0.5N$ (linear: τ_H=0.615, τ_∞=0.223; ungm: τ_H=0.567, τ_∞=0.351; volatility: τ_H=0.587, τ_∞=0.194; bearing: τ_H=0.596, τ_∞=0.285; lorenz63: τ_H=0.599, τ_∞=0.255; lorenz96: τ_H=0.62, τ_∞=0.187). 50 steps; 100 filter runs over 10 observation sequences (10 for OT and Gumbel).

	E_2				E_H (matched)				E_∞ (matched)			
Method	32	128	512	2048	32	128	512	2048	32	128	512	2048
DSR (τ=0.1, paper)	**0.52**	**0.506**	0.505	0.504	0.521	0.507	0.505	0.504	0.522	0.507	**0.504**	**0.503**
DSR (τ=0.1, unsorted)	0.524	0.51	0.509	0.507	0.524	0.51	0.51	0.508	**0.524**	**0.509**	**0.507**	**0.505**
DSR (τ=0.5)	**0.523**	0.509	0.509	0.507	0.523	0.509	0.51	0.508	0.523	**0.508**	**0.507**	**0.506**
DSR (τ=2.5/N)†	0.521	**0.507**	0.504	0.503	**0.52**	0.507	0.504	**0.503**	0.521	0.507	**0.504**	0.503
Systematic	**0.52**	0.506	0.505	**0.503**	0.521	0.507	**0.504**	0.503	0.523	**0.506**	0.504	0.503
Multinomial	0.521	0.507	0.504	**0.503**	**0.52**	0.508	0.505	0.503	0.521	**0.506**	**0.504**	0.503
Systematic, stop-gradient (PyDPF)	**0.521**	0.506	0.505	**0.503**	0.522	0.507	**0.504**	0.503	0.523	**0.506**	0.504	0.503
Soft (α=0.5)	0.523	0.508	0.505	0.503	**0.522**	**0.507**	0.505	**0.503**	0.528	0.508	**0.505**	0.503
OT (PyDPF)	0.53	0.515	–	–	**0.528**	0.514	–	–	0.534	**0.511**	–	–
OT (PFJAX)	0.529	0.515	–	–	**0.529**	0.514	–	–	0.534	**0.512**	–	–
OT (FilterFlow)	0.53	0.515	–	–	**0.53**	0.514	–	–	0.535	**0.511**	–	–
Eigen moment matching	**0.52**	0.507	0.505	0.503	0.52	0.507	**0.505**	**0.503**	0.522	**0.507**	0.505	0.503
Gaussian moment matching	**0.521**	**0.507**	**0.504**	0.503	0.521	0.507	0.504$^-$	**0.503**$^+$	0.522	0.507	0.504	0.503
Gumbel-softmax	0.526	0.514	–	–	**0.524**	0.515	–	–	0.535	**0.511**$^+$	–	–
Variational (KL)	**0.519**	0.507	**0.504**	0.503	0.521$^-$	**0.507**	0.505	**0.503**	0.524$^-$	0.507	0.505	0.503

†Local diagnostic, not from the paper: τ rescaled so that τN equals the paper's value at N=25. Cells are means; $^+/^-$: paired difference from E_2 on the same trials significantly better/worse (95% CI over observation sequences excludes 0). Bold marks the best trigger for that method and N. A dash means not run (dense methods stop at N=128).

Table 29: Linear Gaussian SSM: posterior-mean RMSE vs. Kalman (lower is better), matched resampling rate. Resample when $E_H < \tau_H N$ or $E_\infty < \tau_\infty N$, with per-model thresholds calibrated so that systematic resampling resamples as often as under $E_2 < 0.5N$ (linear: τ_H=0.615, τ_∞=0.223; ungm: τ_H=0.567, τ_∞=0.351; volatility: τ_H=0.587, τ_∞=0.194; bearing: τ_H=0.596, τ_∞=0.285; lorenz63: τ_H=0.599, τ_∞=0.255; lorenz96: τ_H=0.62, τ_∞=0.187). 50 steps; 100 filter runs over 10 observation sequences (10 for OT and Gumbel).

	E_2				E_H (matched)				E_∞ (matched)			
Method	32	128	512	2048	32	128	512	2048	32	128	512	2048
DSR (τ=0.1, paper)	**0.138**	**0.0743**	**0.0442**	0.0339	0.138	0.0743	0.0454	**0.0337**	0.142	0.0755	0.0445	0.034
DSR (τ=0.1, unsorted)	**0.151**	0.1	**0.0803**	0.0745	0.151	**0.0997**	0.0818	**0.0737**	0.151	0.0999	0.0803	0.0751
DSR (τ=0.5)	0.153	0.101	0.0797	0.0732	**0.153**	**0.0998**	0.081	**0.0727**	0.153	0.1	**0.0796**	0.0739
DSR (τ=2.5/N)†	0.136	**0.0668**	0.0326	0.0171	**0.135**	0.0672	**0.0325**	0.0172	0.14	0.0689	0.0329	**0.0167**
Systematic	0.132	0.0675	0.0324	0.017	**0.131**	**0.067**	0.0328	0.0168	0.137	0.0689	**0.0321**	**0.0167**
Multinomial	**0.134**	0.0692	**0.0336**	0.0177	0.135	**0.0688**	0.0339	0.0173	0.139	0.0697	0.0342	**0.0169**
Systematic, stop-gradient (PyDPF)	**0.131**	**0.0678**	0.0329	0.0169	0.132	0.0686	0.0327	**0.0169**	0.136	0.0699	**0.0322**	0.0169
Soft (α=0.5)	0.145	0.0754	0.0357	**0.0182**	**0.142**	**0.0752**	**0.0357**	0.0184	0.153	0.0771	0.037	0.0183
OT (PyDPF)	**0.127**	0.067	–	–	0.128	0.0673	–	–	0.139	**0.0647**	–	–
OT (PFJAX)	**0.127**	0.0672	–	–	0.129	0.0675	–	–	0.139	**0.0646**	–	–
OT (FilterFlow)	**0.128**	0.0679	–	–	0.129	0.0686	–	–	0.139	**0.0655**	–	–
Eigen moment matching	**0.131**	**0.0678**	0.0321	**0.0166**	0.131	0.0682	0.032	0.0166	0.135	0.0688	**0.0319**	0.0167
Gaussian moment matching	**0.133**	0.0707	0.0335	0.0172	0.135	**0.0707**	**0.0335**	0.0173	0.14	0.0723	0.0341	**0.0167**
Gumbel-softmax	**0.133**	0.0793	–	–	0.135	0.0773	–	–	0.144	**0.0769**	–	–
Variational (KL)	**0.132**	0.067	0.0334	**0.0181**	0.133	**0.0668**	**0.0332**	0.0183	0.14	0.0692	0.0338	0.0181

†Local diagnostic, not from the paper: τ rescaled so that τN equals the paper's value at N=25. Cells are means; $^+/^-$: paired difference from E_2 on the same trials significantly better/worse (95% CI over observation sequences excludes 0). Bold marks the best trigger for that method and N. A dash means not run (dense methods stop at N=128).

Table 30: Linear Gaussian SSM: log-likelihood error vs. Kalman (closer to 0 is better), matched resampling rate. Resample when $E_H < \tau_H N$ or $E_\infty < \tau_\infty N$, with per-model thresholds calibrated so that systematic resampling resamples as often as under $E_2 < 0.5N$ (linear: τ_H=0.615, τ_∞=0.223; ungm: τ_H=0.567, τ_∞=0.351; volatility: τ_H=0.587, τ_∞=0.194; bearing: τ_H=0.596, τ_∞=0.285; lorenz63: τ_H=0.599, τ_∞=0.255; lorenz96: τ_H=0.62, τ_∞=0.187). 50 steps; 100 filter runs over 10 observation sequences (10 for OT and Gumbel).

	E_2				E_H (matched)				E_∞ (matched)			
Method	32	128	512	2048	32	128	512	2048	32	128	512	2048
DSR (τ=0.1, paper)	−0.847	−0.19	−0.0486	0.0111	**−0.834**	−0.221	−0.0673	**0.00494**	−0.859	**−0.156**	**−0.0337**	0.0191
DSR (τ=0.1, unsorted)	**−0.854**	−0.265	−0.195	−0.117	−0.898	−0.253	−0.227	−0.122	−0.913	**−0.202**	**−0.123**	**−0.0559**
DSR (τ=0.5)	−0.887	−0.235	−0.195	−0.114	**−0.839**	−0.23	−0.218	−0.133	−0.908	**−0.183**	**−0.115**	**−0.0585**
DSR (τ=2.5/N)†	−0.839	−0.205	**−0.0415**	0.00311	−0.794	−0.205	−0.0518	$\mathbf{1.57 \times 10^{-4}}$	**−0.78**	**−0.155**	−0.0487	−0.00338
Systematic	−0.829	−0.186	−0.0508	0.0258	−0.781	**−0.181**	**−0.0177**	0.0232	**−0.76**	−0.188	−0.0305	$\mathbf{-2.05 \times 10^{-4}}$
Multinomial	−0.903	−0.207	−0.0361	0.0103	−0.876	−0.243	−0.0573	0.00936	**−0.814**	**−0.181**	**−0.0298**	**0.0023**
Systematic, stop-gradient (PyDPF)	−0.756	**−0.171**	**−0.0196**	**−0.00821**	**−0.754**	−0.216	−0.0518	0.0184	−0.804	−0.198	−0.0396	0.0151
Soft (α=0.5)	−0.93	−0.333	**−0.0328**	−0.0124	**−0.837**	**−0.25**	−0.0372	**−0.00892**	−1.22	−0.27	−0.0633	−0.0111
OT (PyDPF)	−1.32	−0.277	–	–	**−1.26**	−0.279	–	–	−1.41	**−0.228**	–	–
OT (PFJAX)	−1.28	−0.276	–	–	**−1.25**	−0.278	–	–	−1.4	**−0.227**	–	–
OT (FilterFlow)	−1.33	−0.269	–	–	**−1.29**	−0.285	–	–	−1.43	**−0.218**	–	–
Eigen moment matching	−0.852	−0.23	**−0.0483**	6.98×10^{-4}	−0.84	−0.257	−0.0558	$\mathbf{-2.92 \times 10^{-4}}$	**−0.811**	**−0.185**	−0.0493	−0.00128
Gaussian moment matching	−0.825	−0.143	−0.064	0.00867	−0.806	**−0.14**	−0.0684	**0.00125**	**−0.776**	−0.156	**−0.0443**	0.00241
Gumbel-softmax	**−1.33**	−0.236	–	–	−1.38	−0.209	–	–	−1.61	**−0.0804**	–	–
Variational (KL)	−0.827	**−0.145**	−0.041	**0.00757**	**−0.777**	−0.152	−0.0497	0.00883	−0.902	−0.159	**−0.0259**	−0.0131

†Local diagnostic, not from the paper: τ rescaled so that τN equals the paper's value at N=25. Cells are means; $^+/^-$: paired difference from E_2 on the same trials significantly better/worse (95% CI over observation sequences excludes 0). Bold marks the best trigger for that method and N. A dash means not run (dense methods stop at N=128).

Table 31: Linear Gaussian SSM: 95% coverage (closer to 0.95 is better), matched resampling rate. Resample when $E_H < \tau_H N$ or $E_\infty < \tau_\infty N$, with per-model thresholds calibrated so that systematic resampling resamples as often as under $E_2 < 0.5N$ (linear: τ_H=0.615, τ_∞=0.223; ungm: τ_H=0.567, τ_∞=0.351; volatility: τ_H=0.587, τ_∞=0.194; bearing: τ_H=0.596, τ_∞=0.285; lorenz63: τ_H=0.599, τ_∞=0.255; lorenz96: τ_H=0.62, τ_∞=0.187). 50 steps; 100 filter runs over 10 observation sequences (10 for OT and Gumbel).

	E_2				E_H (matched)				E_∞ (matched)			
Method	32	128	512	2048	32	128	512	2048	32	128	512	2048
DSR (τ=0.1, paper)	0.903	0.923	0.926	**0.928**	0.903	0.923	0.926	0.927	**0.904**	**0.924**	**0.927**	0.926
DSR (τ=0.1, unsorted)	0.895	0.913	0.916	0.917	**0.895**	0.912	0.915	0.916	0.894	**0.914**	**0.92**	**0.919**
DSR (τ=0.5)	0.894	0.912	0.916	0.917	**0.894**	0.912	0.916	0.916	0.892	**0.914**	**0.92**	**0.919**
DSR (τ=2.5/N)†	0.904	**0.929**	**0.931**	0.933	0.905	**0.929**	**0.931**	**0.933**	**0.906**	**0.929**$^-$	0.93	0.933
Systematic	0.912	0.929	0.931	**0.933**	**0.912**	**0.93**	0.93	**0.933**	0.908	0.929	**0.932**	0.932
Multinomial	0.911	0.928	**0.931**	0.933	**0.911**	0.928	0.931	0.932	0.908	**0.93**	**0.931**	**0.933**
Systematic, stop-gradient (PyDPF)	**0.909**	**0.931**	0.931	0.932	0.909	0.929	0.93	0.932	0.909	0.928	**0.933**	**0.933**
Soft (α=0.5)	**0.908**	**0.927**	0.931	0.932	0.905	0.927	0.931	**0.933**	0.902	0.924	**0.932**	0.932
OT (PyDPF)	0.916	**0.928**	–	–	**0.918**	**0.928**	–	–	0.91	**0.928**	–	–
OT (PFJAX)	**0.918**	**0.93**	–	–	**0.918**	**0.93**	–	–	0.91	0.928	–	–
OT (FilterFlow)	0.916	**0.924**	–	–	**0.918**	**0.924**	–	–	0.91	**0.924**	–	–
Eigen moment matching	**0.914**	0.929	**0.931**	**0.932**	0.913	**0.929**	0.931	**0.932**	0.912	0.928	0.931	0.932
Gaussian moment matching	0.911	**0.929**	0.931	0.934	**0.912**	0.929	0.932	**0.934**	0.905$^-$	0.928	**0.932**	**0.934**
Gumbel-softmax	0.902	**0.928**	–	–	**0.904**	**0.928**	–	–	0.888	0.922	–	–
Variational (KL)	**0.909**	**0.929**	0.93	0.932	0.908	0.928	0.931	**0.933**	0.905	0.928	**0.932**$^+$	0.932

†Local diagnostic, not from the paper: τ rescaled so that τN equals the paper's value at N=25. Cells are means; $^+/^-$: paired difference from E_2 on the same trials significantly better/worse (95% CI over observation sequences excludes 0). Bold marks the best trigger for that method and N. A dash means not run (dense methods stop at N=128).

Table 32: Linear Gaussian SSM: predictive NLL (lower is better), matched resampling rate. Resample when $E_H < \tau_H N$ or $E_\infty < \tau_\infty N$, with per-model thresholds calibrated so that systematic resampling resamples as often as under $E_2 < 0.5N$ (linear: τ_H=0.615, τ_∞=0.223; ungm: τ_H=0.567, τ_∞=0.351; volatility: τ_H=0.587, τ_∞=0.194; bearing: τ_H=0.596, τ_∞=0.285; lorenz63: τ_H=0.599, τ_∞=0.255; lorenz96: τ_H=0.62, τ_∞=0.187). 50 steps; 100 filter runs over 10 observation sequences (10 for OT and Gumbel).

	E_2				E_H (matched)				E_∞ (matched)			
Method	32	128	512	2048	32	128	512	2048	32	128	512	2048
DSR (τ=0.1, paper)	1.51	1.5	1.49	1.49	**1.51**	1.5	1.49	1.49	1.51	**1.5**	**1.49**	**1.49**
DSR (τ=0.1, unsorted)	**1.51**	1.5	1.5	1.5	1.51	1.5	1.5	1.5	1.51	**1.5**	**1.5**	**1.49**
DSR (τ=0.5)	1.51	1.5	1.5	1.5	**1.51**	1.5	1.5	1.5	1.51	**1.5**	**1.5**	**1.49**
DSR (τ=2.5/N)†	1.51	1.5	**1.49**	**1.49**	1.51	1.5	1.49	1.49	**1.51**	**1.5**$^+$	1.49	1.49
Systematic	1.51	1.5	1.49	**1.49**	1.51	**1.5**	**1.49**	1.49	**1.51**	1.5	1.49	1.49
Multinomial	1.51	1.5	1.49	**1.49**	1.51	1.5	1.49	1.49	**1.51**	**1.5**	**1.49**	1.49
Systematic, stop-gradient (PyDPF)	1.51	**1.5**	**1.49**	1.49	**1.51**	1.5	1.49	**1.49**	1.51	1.5	1.49	1.49
Soft (α=0.5)	1.51	1.5	**1.49**	1.49	**1.51**	**1.5**	1.49	**1.49**	1.52$^-$	1.5	1.49	1.49
OT (PyDPF)	1.52	1.5	–	–	**1.52**	1.5	–	–	1.52	**1.5**	–	–
OT (PFJAX)	1.52	1.5	–	–	**1.52**	1.5	–	–	1.52	**1.5**	–	–
OT (FilterFlow)	1.52	1.5	–	–	**1.52**	1.5	–	–	1.52	**1.5**	–	–
Eigen moment matching	1.51	1.5	**1.49**	**1.49**	1.51	1.5	1.49	1.49	**1.51**	**1.5**	1.49	1.49
Gaussian moment matching	1.51	1.5	1.49	**1.49**	1.51	**1.5**	1.49	1.49	**1.51**	1.5	**1.49**	1.49
Gumbel-softmax	**1.52**	1.5	–	–	1.52	1.5	–	–	1.52	**1.49**	–	–
Variational (KL)	1.51	**1.5**	1.49	1.49	**1.51**	1.5	1.49	**1.49**	1.51	1.5	**1.49**	1.49

†Local diagnostic, not from the paper: τ rescaled so that τN equals the paper's value at N=25. Cells are means; $^+/^-$: paired difference from E_2 on the same trials significantly better/worse (95% CI over observation sequences excludes 0). Bold marks the best trigger for that method and N. A dash means not run (dense methods stop at N=128).

Table 33: Linear Gaussian SSM: resampling steps out of 50, matched resampling rate. Resample when $E_H < \tau_H N$ or $E_\infty < \tau_\infty N$, with per-model thresholds calibrated so that systematic resampling resamples as often as under $E_2 < 0.5N$ (linear: τ_H=0.615, τ_∞=0.223; ungm: τ_H=0.567, τ_∞=0.351; volatility: τ_H=0.587, τ_∞=0.194; bearing: τ_H=0.596, τ_∞=0.285; lorenz63: τ_H=0.599, τ_∞=0.255; lorenz96: τ_H=0.62, τ_∞=0.187). 50 steps; 100 filter runs over 10 observation sequences (10 for OT and Gumbel).

	E_2				E_H (matched)				E_∞ (matched)			
Method	32	128	512	2048	32	128	512	2048	32	128	512	2048
DSR (τ=0.1, paper)	15.5	16.2	16.3	16.2	15.6	16.2	16.3	16.1	13.5	15.3	16.5	16.9
DSR (τ=0.1, unsorted)	15.1	15.6	15.8	15.7	15.2	15.3	15.6	15.4	13.3	15	16.2	16.4
DSR (τ=0.5)	15	15.6	15.8	15.7	14.9	15.3	15.5	15.4	13.1	15	16.2	16.4
DSR (τ=2.5/N)†	15.6	16.5	16.5	16.4	15.7	16.6	16.6	16.6	13.5	15.5	16.7	17.1
Systematic	16.1	16.5	16.5	16.5	16.1	16.4	16.6	16.7	13.7	15.6	16.7	17.1
Multinomial	15.8	16.5	16.5	16.5	15.9	16.4	16.6	16.7	13.7	15.4	16.6	17.1
Systematic, stop-gradient (PyDPF)	15.9	16.4	16.6	16.4	15.9	16.4	16.6	16.6	13.5	15.6	16.7	17.2
Soft (α=0.5)	22.7	23.3	23.6	23.6	23.7	24.5	24.8	25	17.5	20.5	21.9	22.6
OT (PyDPF)	15.9	16.6	–	–	15.9	16.7	–	–	13.5	16	–	–
OT (PFJAX)	15.8	16.6	–	–	15.9	16.7	–	–	13.5	16	–	–
OT (FilterFlow)	15.9	16.6	–	–	16	16.7	–	–	13.5	16	–	–
Eigen moment matching	16	16.4	16.5	16.5	16.2	16.4	16.6	16.6	13.5	15.6	16.6	17.1
Gaussian moment matching	16	16.4	16.6	16.5	16.1	16.4	16.5	16.6	13.6	15.5	16.6	17.1
Gumbel-softmax	15.8	16.1	–	–	15.7	16.1	–	–	13.5	15.4	–	–
Variational (KL)	15.8	16.5	16.6	16.5	16	16.4	16.6	16.6	13.5	15.6	16.6	17.1

†Local diagnostic, not from the paper: τ rescaled so that τN equals the paper's value at N=25. Cells are means; $^+/^-$: paired difference from E_2 on the same trials significantly better/worse (95% CI over observation sequences excludes 0). Bold marks the best trigger for that method and N. A dash means not run (dense methods stop at N=128).

Table 34: UNGM: state RMSE (lower is better), matched resampling rate. Resample when $E_H < \tau_H N$ or $E_\infty < \tau_\infty N$, with per-model thresholds calibrated so that systematic resampling resamples as often as under $E_2 < 0.5N$ (linear: τ_H=0.615, τ_∞=0.223; ungm: τ_H=0.567, τ_∞=0.351; volatility: τ_H=0.587, τ_∞=0.194; bearing: τ_H=0.596, τ_∞=0.285; lorenz63: τ_H=0.599, τ_∞=0.255; lorenz96: τ_H=0.62, τ_∞=0.187). 50 steps; 100 filter runs over 10 observation sequences (10 for OT and Gumbel).

	E_2				E_H (matched)				E_∞ (matched)			
Method	32	128	512	2048	32	128	512	2048	32	128	512	2048
DSR (τ=0.1, paper)	6.22	**5.15**	4.79	4.78	6.17	5.19	**4.6**	**4.66**	**6.12**	5.21	4.81	4.78
DSR (τ=0.1, unsorted)	**7.52**	**7.63**	**8.64**	**8.68**	7.68$^-$	7.74	8.86	9.01	7.64	7.96	8.78	9
DSR (τ=0.5)	8.64	**8.52**	**8.81**	**8.88**	8.69	8.55	9.09	9.04	**8.58**	8.77	8.87	9.02
DSR (τ=2.5/N)†	6.02	4.98	4.56	4.52	6.06	**4.97**	**4.53**	4.52	**6.01**	5.03	4.59	**4.51**
Systematic	**6.01**	4.91	4.49	**4.48**	6.11	4.9	**4.49**	4.51	6.05	**4.83**	4.61	4.51
Multinomial	5.97	4.92	4.57	4.52	5.86$^+$	4.93	**4.51**	**4.5**	**5.83**$^+$	**4.85**	4.56	4.52
Systematic, stop-gradient (PyDPF)	**5.83**	4.88	4.51	4.54	5.9	**4.85**$^+$	**4.5**	4.53	5.92	4.86	4.57	**4.51**
Soft (α=0.5)	**5.42**	4.83	4.54	**4.44**	5.61$^-$	4.82	**4.47**	4.46	5.6	**4.81**	4.56	4.46
OT (PyDPF)	5.6	5.42	–	–	**5.33**	5.42	–	–	6.1	**5.38**	–	–
OT (PFJAX)	6.1	5.33	–	–	**5.96**	5.35	–	–	6.44	**5.33**	–	–
OT (FilterFlow)	6.01	5.48	–	–	6.05	5.47	–	–	**5.72**	**5.38**$^+$	–	–
Eigen moment matching	**5.99**	5.17	4.98	4.83	6.06	**5.15**	**4.93**	**4.78**	6.04	5.26	5.03	4.87
Gaussian moment matching	6.02	**4.94**	4.8	4.66	**5.97**	4.95	**4.72**	**4.62**	6.07	5.01	4.83	4.69
Gumbel-softmax	6.51	5.27	–	–	**6.39**	5.35	–	–	6.51	**5.15**	–	–
Variational (KL)	**5.79**	**4.83**	4.56	**4.48**	5.81	4.93	**4.53**	4.53	5.84	4.85	4.61	4.51

†Local diagnostic, not from the paper: τ rescaled so that τN equals the paper's value at N=25. Cells are means; $^+/^-$: paired difference from E_2 on the same trials significantly better/worse (95% CI over observation sequences excludes 0). Bold marks the best trigger for that method and N. A dash means not run (dense methods stop at N=128).

Table 35: UNGM: 95% coverage (closer to 0.95 is better), matched resampling rate. Resample when $E_H < \tau_H N$ or $E_\infty < \tau_\infty N$, with per-model thresholds calibrated so that systematic resampling resamples as often as under $E_2 < 0.5N$ (linear: τ_H=0.615, τ_∞=0.223; ungm: τ_H=0.567, τ_∞=0.351; volatility: τ_H=0.587, τ_∞=0.194; bearing: τ_H=0.596, τ_∞=0.285; lorenz63: τ_H=0.599, τ_∞=0.255; lorenz96: τ_H=0.62, τ_∞=0.187). 50 steps; 100 filter runs over 10 observation sequences (10 for OT and Gumbel).

	E_2				E_H (matched)				E_∞ (matched)			
Method	32	128	512	2048	32	128	512	2048	32	128	512	2048
DSR (τ=0.1, paper)	0.813	**0.926**	0.955	0.961	0.813	0.923	0.96	0.964	**0.817**	0.925	**0.954**	**0.96**
DSR (τ=0.1, unsorted)	**0.737**	**0.784**	**0.763**	**0.766**	0.734	0.778	0.751	0.747	0.73	0.762^-	0.749	0.749
DSR (τ=0.5)	0.672	**0.738**	**0.752**	**0.768**	0.668	0.731	0.736	0.756	**0.673**	0.717	0.744	0.759
DSR (τ=2.5/N)†	0.822	0.929	0.963	0.971	**0.822**	**0.93**	0.965	**0.971**	**0.822**	0.927	**0.962**	**0.971**
Systematic	**0.821**	0.93	0.966	0.971	0.816	0.93	0.966	**0.971**	0.82	**0.933**	**0.964**	**0.971**
Multinomial	0.818	0.928	**0.964**	**0.972**	**0.823**	0.931	0.965	0.972	0.822	**0.931**	0.965	0.972
Systematic, stop-gradient (PyDPF)	**0.828**	0.932	0.966	0.972	0.825	0.931	0.967	**0.971**	0.825	$\mathbf{0.933}^+$	**0.965**	0.972
Soft (α=0.5)	**0.797**	0.928	**0.962**	**0.97**	0.791	**0.929**	0.962	0.971	0.796	0.925	0.962	**0.97**
OT (PyDPF)	0.814	**0.914**	–	–	**0.83**	0.91	–	–	0.794	**0.914**	–	–
OT (PFJAX)	0.8	**0.918**	–	–	**0.808**	0.916	–	–	0.794	0.916	–	–
OT (FilterFlow)	0.806	0.908	–	–	**0.81**	**0.916**	–	–	0.804	0.91	–	–
Eigen moment matching	**0.828**	0.93	0.953	0.963	0.824	**0.932**	0.954	**0.962**	0.823	0.926	**0.952**	0.963
Gaussian moment matching	0.824	0.933	0.962	0.97	**0.825**	**0.934**	0.964	0.971	0.822	0.932	**0.96**	**0.969**
Gumbel-softmax	**0.82**	**0.918**	–	–	0.81	**0.918**	–	–	0.812	0.916	–	–
Variational (KL)	0.825	0.93	0.963	0.971	**0.827**	0.927	0.965	**0.97**	0.824	**0.93**	**0.962**	**0.97**

†Local diagnostic, not from the paper: τ rescaled so that τN equals the paper's value at N=25. Cells are means; $^+/^-$: paired difference from E_2 on the same trials significantly better/worse (95% CI over observation sequences excludes 0). Bold marks the best trigger for that method and N. A dash means not run (dense methods stop at N=128).

Table 36: UNGM: predictive NLL (lower is better), matched resampling rate. Resample when $E_H < \tau_H N$ or $E_\infty < \tau_\infty N$, with per-model thresholds calibrated so that systematic resampling resamples as often as under $E_2 < 0.5N$ (linear: τ_H=0.615, τ_∞=0.223; ungm: τ_H=0.567, τ_∞=0.351; volatility: τ_H=0.587, τ_∞=0.194; bearing: τ_H=0.596, τ_∞=0.285; lorenz63: τ_H=0.599, τ_∞=0.255; lorenz96: τ_H=0.62, τ_∞=0.187). 50 steps; 100 filter runs over 10 observation sequences (10 for OT and Gumbel).

	E_2				E_H (matched)				E_∞ (matched)			
Method	32	128	512	2048	32	128	512	2048	32	128	512	2048
DSR (τ=0.1, paper)	5.53	**3.23**	2.86	2.77	**5.37**	3.27	**2.82**	**2.76**	5.39	3.32	2.86	2.77
DSR (τ=0.1, unsorted)	**7.21**	**6.09**	6.2	**5.57**	7.56	6.15	6.27	5.73	7.32	6.2	**6.15**	5.79
DSR (τ=0.5)	9.33	7.06	6.21	5.46	9.33	7.06	6.44	5.62	$\mathbf{9.13}^+$	**7.06**	**6.16**	**5.45**
DSR (τ=2.5/N)†	5.33	3.19	2.8	2.7	5.39	**3.17**	**2.79**	2.7	**5.27**	3.23	2.82	**2.7**
Systematic	5.36	3.18	**2.78**	2.7	5.43	3.18	2.79	**2.7**	**5.31**	**3.16**	2.82	2.7
Multinomial	5.26	3.3	2.82	**2.7**	$\mathbf{5.08}^+$	3.32	**2.79**	2.7	5.12^+	**3.25**	2.82	2.7
Systematic, stop-gradient (PyDPF)	**4.99**	3.14	2.79	2.7	5.05	**3.13**	**2.78**	2.7	5	3.16	2.81	**2.7**
Soft (α=0.5)	**4.32**	3.13	2.81	2.71	4.43	**3.08**	**2.81**	2.72	4.34	3.15	2.82	**2.71**
OT (PyDPF)	4.18	3.18	–	–	**3.63**	3.18	–	–	5.48	**3.18**	–	–
OT (PFJAX)	4.6	**3.19**	–	–	**4.55**	3.19	–	–	5.87	3.19	–	–
OT (FilterFlow)	4.53	3.2	–	–	4.6	3.2	–	–	**4.29**	**3.2**	–	–
Eigen moment matching	**4.98**	3.17	2.88	2.77	5	**3.16**	**2.86**	**2.76**	5.03	3.29	2.89	2.76
Gaussian moment matching	5.06	**3.13**	2.82	2.71	**5.04**	3.15	**2.8**	**2.71**	5.1	3.17	2.84	2.72
Gumbel-softmax	4.82	**3.16**	–	–	**4.8**	3.16	–	–	5.37	3.16	–	–
Variational (KL)	4.98	**3.18**	2.83	2.71	**4.87**	3.24	**2.82**	2.71	4.95	3.2	2.85	**2.7**

†Local diagnostic, not from the paper: τ rescaled so that τN equals the paper's value at N=25. Cells are means; $^+/^-$: paired difference from E_2 on the same trials significantly better/worse (95% CI over observation sequences excludes 0). Bold marks the best trigger for that method and N. A dash means not run (dense methods stop at N=128).

Table 37: UNGM: resampling steps out of 50, matched resampling rate. Resample when $E_H < \tau_H N$ or $E_\infty < \tau_\infty N$, with per-model thresholds calibrated so that systematic resampling resamples as often as under $E_2 < 0.5N$ (linear: τ_H=0.615, τ_∞=0.223; ungm: τ_H=0.567, τ_∞=0.351; volatility: τ_H=0.587, τ_∞=0.194; bearing: τ_H=0.596, τ_∞=0.285; lorenz63: τ_H=0.599, τ_∞=0.255; lorenz96: τ_H=0.62, τ_∞=0.187). 50 steps; 100 filter runs over 10 observation sequences (10 for OT and Gumbel).

	E_2				E_H (matched)				E_∞ (matched)			
Method	32	128	512	2048	32	128	512	2048	32	128	512	2048
DSR (τ=0.1, paper)	37	37.4	37.7	37.8	37.1	37.8	37.9	38.1	37.1	38	37.8	38
DSR (τ=0.1, unsorted)	37.2	37.5	37.5	37.5	37.4	37.5	37.6	37.7	37.4	38	38.1	38.2
DSR (τ=0.5)	37.4	37.5	37.7	37.9	37.3	37.6	37.9	37.9	37.7	38.3	38.3	38.5
DSR (τ=2.5/N)†	37	37.3	37.3	37.2	37.1	37.7	37.5	37.6	37.2	37.7	37.6	37.6
Systematic	36.9	37.1	37.3	37.2	37	37.4	37.6	37.6	37.2	37.6	37.6	37.6
Multinomial	36.7	37.3	37.3	37.3	36.9	37.5	37.7	37.6	37	37.6	37.7	37.6
Systematic, stop-gradient (PyDPF)	36.8	37.2	37.3	37.3	37.1	37.4	37.6	37.5	36.9	37.6	37.6	37.5
Soft (α=0.5)	44.1	44.1	43.9	43.9	45.2	45.6	45.5	45.6	42.7	43.2	43.3	43.2
OT (PyDPF)	37	37.4	–	–	37.5	37.4	–	–	37.5	38	–	–
OT (PFJAX)	37.2	37.1	–	–	37.4	37.4	–	–	37.8	37.9	–	–
OT (FilterFlow)	37	37.6	–	–	37.2	37.5	–	–	37.4	38.2	–	–
Eigen moment matching	37.5	38.1	38.4	38.6	37.8	38.6	38.7	38.9	37.6	38.1	38.2	38.4
Gaussian moment matching	37.5	37.9	38.5	38.5	37.8	38.2	38.6	38.8	37.6	38.1	38.4	38.5
Gumbel-softmax	37.6	37.6	–	–	37.2	37.5	–	–	37.7	38	–	–
Variational (KL)	36.8	37.1	37.4	37.2	37.1	37.5	37.6	37.5	37	37.5	37.6	37.5

†Local diagnostic, not from the paper: τ rescaled so that τN equals the paper's value at N=25. Cells are means; $^+/^-$: paired difference from E_2 on the same trials significantly better/worse (95% CI over observation sequences excludes 0). Bold marks the best trigger for that method and N. A dash means not run (dense methods stop at N=128).

Table 38: Stochastic volatility: state RMSE (lower is better), matched resampling rate. Resample when $E_H < \tau_H N$ or $E_\infty < \tau_\infty N$, with per-model thresholds calibrated so that systematic resampling resamples as often as under $E_2 < 0.5N$ (linear: τ_H=0.615, τ_∞=0.223; ungm: τ_H=0.567, τ_∞=0.351; volatility: τ_H=0.587, τ_∞=0.194; bearing: τ_H=0.596, τ_∞=0.285; lorenz63: τ_H=0.599, τ_∞=0.255; lorenz96: τ_H=0.62, τ_∞=0.187). 50 steps; 100 filter runs over 10 observation sequences (10 for OT and Gumbel).

	E_2				E_H (matched)				E_∞ (matched)			
Method	32	128	512	2048	32	128	512	2048	32	128	512	2048
DSR (τ=0.1, paper)	**0.425**	**0.413**	0.412	0.411	0.428	0.416	0.415	0.414	0.428	0.413	**0.41**	**0.41**
DSR (τ=0.1, unsorted)	0.431	0.423	0.418	0.418	0.431	0.426	0.425	0.427	**0.425**	**0.414**	**0.412**	**0.415**
DSR (τ=0.5)	0.431	0.423	0.418	0.418	0.432	0.424	0.425	0.425	**0.427**	**0.415**	**0.413**	**0.415**
DSR (τ=2.5/N)†	**0.425**	**0.415**	**0.415**	0.414	0.428	0.415	0.415	**0.414**	0.429	0.416	0.415	0.414
Systematic	**0.428**	**0.417**	0.415	**0.414**	0.431^-	0.418	**0.415**	0.414	0.431	0.417	0.416	0.414
Multinomial	0.43	0.417	0.415	0.414	**0.429**	**0.417**	**0.415**	**0.414**	0.432	0.418	0.416	0.414
Systematic, stop-gradient (PyDPF)	**0.427**	0.416	**0.414**	0.414	0.428	**0.416**	0.415^-	**0.414**	0.429	0.417	0.416^-	0.414
Soft (α=0.5)	**0.429**	0.419	0.415	0.415	0.429	0.42	**0.414**	**0.414**	0.436^-	**0.417**	0.416	0.414
OT (PyDPF)	0.426	**0.418**	–	–	0.427	0.42	–	–	**0.425**	0.423	–	–
OT (PFJAX)	0.426	**0.418**	–	–	0.427	0.42	–	–	**0.425**	0.423	–	–
OT (FilterFlow)	0.427	**0.417**	–	–	0.428	0.42	–	–	**0.424**	0.423	–	–
Eigen moment matching	0.428	**0.416**	**0.415**	0.414	**0.427**	0.417	0.415	0.414	0.428	0.419	0.415	**0.414**
Gaussian moment matching	**0.426**	0.416	0.415	**0.414**	0.427	**0.415**	**0.415**	0.414	0.426	0.416	0.416	0.414
Gumbel-softmax	0.427	**0.415**	–	–	**0.421**	0.422	–	–	0.424	0.421	–	–
Variational (KL)	**0.426**	**0.417**	**0.413**	0.413	0.428	0.419	0.415^-	0.414	0.431^-	0.418	0.414	**0.412**

†Local diagnostic, not from the paper: τ rescaled so that τN equals the paper's value at N=25. Cells are means; $^+/^-$: paired difference from E_2 on the same trials significantly better/worse (95% CI over observation sequences excludes 0). Bold marks the best trigger for that method and N. A dash means not run (dense methods stop at N=128).

Table 39: Stochastic volatility: 95% coverage (closer to 0.95 is better), matched resampling rate. Resample when $E_H < \tau_H N$ or $E_\infty < \tau_\infty N$, with per-model thresholds calibrated so that systematic resampling resamples as often as under $E_2 < 0.5N$ (linear: τ_H=0.615, τ_∞=0.223; ungm: τ_H=0.567, τ_∞=0.351; volatility: τ_H=0.587, τ_∞=0.194; bearing: τ_H=0.596, τ_∞=0.285; lorenz63: τ_H=0.599, τ_∞=0.255; lorenz96: τ_H=0.62, τ_∞=0.187). 50 steps; 100 filter runs over 10 observation sequences (10 for OT and Gumbel).

	E_2				E_H (matched)				E_∞ (matched)			
Method	32	128	512	2048	32	128	512	2048	32	128	512	2048
DSR (τ=0.1, paper)	**0.926**	0.947	0.954	**0.952**	0.923	0.947	**0.952**	0.952^+	0.926	**0.949**	0.955	0.953
DSR (τ=0.1, unsorted)	0.886	0.896	0.905	0.904	0.893	0.904^+	0.907	**0.907**	**0.906**	**0.909**	**0.907**	0.906
DSR (τ=0.5)	0.88	0.893	0.907	0.908	0.883	0.905^+	**0.908**	**0.911**	**0.895**	**0.912**	0.907	0.909
DSR (τ=2.5/N)†	**0.931**	**0.962**	0.975	0.976	0.927	**0.962**	**0.973**	**0.976**	0.929	0.962	0.974	0.977
Systematic	**0.941**	0.966	0.974	0.976	0.938	**0.964**	0.974	**0.975**	0.938	**0.964**	**0.972**	0.976
Multinomial	**0.942**	0.961	0.974	**0.975**	0.938	0.961	**0.972**	0.975	0.933	**0.959**	0.972	0.977
Systematic, stop-gradient (PyDPF)	**0.942**	0.967	0.976	0.977	0.941	0.968	**0.974**	**0.976**	0.937	$\mathbf{0.963}^+$	**0.974**	0.976
Soft (α=0.5)	**0.937**	**0.959**	**0.973**	0.977	0.936	0.96	0.975	**0.976**	0.931	0.965	0.975	0.977
OT (PyDPF)	0.94	0.97	–	–	0.94	0.968	–	–	**0.956**	**0.958**	–	–
OT (PFJAX)	0.938	0.97	–	–	0.94	0.968	–	–	**0.956**	**0.956**	–	–
OT (FilterFlow)	0.936	0.968	–	–	0.934	0.966	–	–	**0.956**	**0.954**	–	–
Eigen moment matching	0.937	0.966	0.975	0.978	0.937	0.966	0.975	0.978	**0.938**	**0.963**	**0.974**	**0.978**
Gaussian moment matching	**0.939**	0.965	0.977	0.978	0.938	**0.964**	**0.975**	**0.976**	0.937	0.965	0.975	0.977
Gumbel-softmax	0.916	**0.946**	–	–	0.926	**0.946**	–	–	**0.948**	0.936	–	–
Variational (KL)	**0.938**	0.96	0.972	0.973	0.935	0.962	0.972	**0.971**	0.934	**0.96**	**0.971**	0.971

†Local diagnostic, not from the paper: τ rescaled so that τN equals the paper's value at N=25. Cells are means; $^+/^-$: paired difference from E_2 on the same trials significantly better/worse (95% CI over observation sequences excludes 0). Bold marks the best trigger for that method and N. A dash means not run (dense methods stop at N=128).

Table 40: Stochastic volatility: predictive NLL (lower is better), matched resampling rate. Resample when $E_H < \tau_H N$ or $E_\infty < \tau_\infty N$, with per-model thresholds calibrated so that systematic resampling resamples as often as under $E_2 < 0.5N$ (linear: τ_H=0.615, τ_∞=0.223; ungm: τ_H=0.567, τ_∞=0.351; volatility: τ_H=0.587, τ_∞=0.194; bearing: τ_H=0.596, τ_∞=0.285; lorenz63: τ_H=0.599, τ_∞=0.255; lorenz96: τ_H=0.62, τ_∞=0.187). 50 steps; 100 filter runs over 10 observation sequences (10 for OT and Gumbel).

	E_2				E_H (matched)				E_∞ (matched)			
Method	32	128	512	2048	32	128	512	2048	32	128	512	2048
DSR (τ=0.1, paper)	1.66	1.66	1.66	**1.66**	1.66	1.66	1.66	1.66	**1.66**	**1.66**	**1.66**	1.66
DSR (τ=0.1, unsorted)	1.67	1.66	**1.66**	**1.66**	1.67	1.66	1.66	1.66	**1.66**	**1.66**	1.66	1.66
DSR (τ=0.5)	1.67	1.66	**1.66**	**1.66**	1.67	1.66	1.66	1.66	**1.66**	**1.66**	1.66	1.66
DSR (τ=2.5/N)†	1.66	1.66	1.66	1.66	1.66	1.66	**1.66**	**1.66**	**1.66**	**1.66**	1.66	1.66
Systematic	**1.66**	1.66	1.66	**1.66**	1.66	1.66	**1.66**	1.66	1.66	**1.66**	1.66	1.66
Multinomial	1.66	1.66	1.66	**1.66**	**1.66**	1.66	$\mathbf{1.66}^+$	1.66	1.66	**1.66**	1.66	1.66^-
Systematic, stop-gradient (PyDPF)	**1.66**	1.66	1.66	**1.66**	1.66	1.66	**1.66**	1.66	1.66	**1.66**	1.66	1.66
Soft (α=0.5)	**1.66**	1.66	1.66	**1.66**	1.66	1.66	**1.66**	1.66	**1.66**	**1.66**	1.66	1.66
OT (PyDPF)	**1.66**	1.66	–	–	1.66	**1.66**	–	–	1.66	1.66	–	–
OT (PFJAX)	**1.66**	1.66	–	–	1.66	**1.66**	–	–	1.66	1.66	–	–
OT (FilterFlow)	**1.66**	1.66	–	–	1.66	**1.66**	–	–	1.66	1.66	–	–
Eigen moment matching	1.66	1.66	**1.66**	**1.66**	1.66	1.66	**1.66**	1.66	**1.66**	**1.66**	1.66	1.66
Gaussian moment matching	**1.66**	1.66	1.66	1.66	1.66	**1.66**	**1.66**	**1.66**	1.66	1.66	1.66	1.66
Gumbel-softmax	**1.66**	1.66	–	–	1.66	**1.66**	–	–	1.66	1.66	–	–
Variational (KL)	1.66	1.66	**1.66**	1.66	**1.66**	1.66	1.66	1.66	1.66	**1.66**	1.66	**1.66**

†Local diagnostic, not from the paper: τ rescaled so that τN equals the paper's value at N=25. Cells are means; $^+/^-$: paired difference from E_2 on the same trials significantly better/worse (95% CI over observation sequences excludes 0). Bold marks the best trigger for that method and N. A dash means not run (dense methods stop at N=128).

Table 41: Stochastic volatility: resampling steps out of 50, matched resampling rate. Resample when $E_H < \tau_H N$ or $E_\infty < \tau_\infty N$, with per-model thresholds calibrated so that systematic resampling resamples as often as under $E_2 < 0.5N$ (linear: τ_H=0.615, τ_∞=0.223; ungm: τ_H=0.567, τ_∞=0.351; volatility: τ_H=0.587, τ_∞=0.194; bearing: τ_H=0.596, τ_∞=0.285; lorenz63: τ_H=0.599, τ_∞=0.255; lorenz96: τ_H=0.62, τ_∞=0.187). 50 steps; 100 filter runs over 10 observation sequences (10 for OT and Gumbel).

	E_2				E_H (matched)				E_∞ (matched)			
Method	32	128	512	2048	32	128	512	2048	32	128	512	2048
DSR (τ=0.1, paper)	3.89	4.16	4.27	4.27	3.51	3.56	3.57	3.59	2.77	3.7	4.46	5.18
DSR (τ=0.1, unsorted)	3.7	3.79	3.91	3.93	3.29	3.42	3.38	3.44	2.74	3.65	4.35	4.94
DSR (τ=0.5)	3.63	3.81	3.92	3.94	3.28	3.39	3.38	3.43	2.74	3.61	4.33	4.94
DSR (τ=2.5/N)†	3.91	4.22	4.27	4.26	3.54	3.76	3.76	3.85	2.83	3.83	4.66	5.54
Systematic	4	4.25	4.24	4.31	3.58	3.69	3.85	3.83	2.9	3.94	4.75	5.44
Multinomial	3.9	4.27	4.35	4.27	3.63	3.74	3.79	3.82	2.89	3.81	4.75	5.49
Systematic, stop-gradient (PyDPF)	4.03	4.28	4.29	4.27	3.59	3.74	3.75	3.84	2.92	3.88	4.73	5.47
Soft (α=0.5)	5.63	5.71	5.8	5.87	5.3	5.31	5.3	5.27	3.66	4.93	6.01	6.86
OT (PyDPF)	4.1	4.2	–	–	3.6	3.5	–	–	2.6	4.2	–	–
OT (PFJAX)	4.1	4.3	–	–	3.6	3.5	–	–	2.6	4.2	–	–
OT (FilterFlow)	4.1	4.3	–	–	3.4	3.5	–	–	2.6	4.2	–	–
Eigen moment matching	4.02	4.26	4.29	4.28	3.58	3.74	3.78	3.86	2.82	3.85	4.81	5.52
Gaussian moment matching	3.97	4.15	4.31	4.24	3.64	3.76	3.75	3.83	2.92	3.93	4.8	5.56
Gumbel-softmax	3.5	4.4	–	–	3.2	3.5	–	–	2.6	4.3	–	–
Variational (KL)	3.97	4.24	4.25	4.27	3.65	3.68	3.72	3.79	2.93	3.94	4.62	5.45

†Local diagnostic, not from the paper: τ rescaled so that τN equals the paper's value at N=25. Cells are means; $^+/^-$: paired difference from E_2 on the same trials significantly better/worse (95% CI over observation sequences excludes 0). Bold marks the best trigger for that method and N. A dash means not run (dense methods stop at N=128).

Table 42: Bearing-only tracking: state RMSE (lower is better), matched resampling rate. Resample when $E_H < \tau_H N$ or $E_\infty < \tau_\infty N$, with per-model thresholds calibrated so that systematic resampling resamples as often as under $E_2 < 0.5N$ (linear: τ_H=0.615, τ_∞=0.223; ungm: τ_H=0.567, τ_∞=0.351; volatility: τ_H=0.587, τ_∞=0.194; bearing: τ_H=0.596, τ_∞=0.285; lorenz63: τ_H=0.599, τ_∞=0.255; lorenz96: τ_H=0.62, τ_∞=0.187). 50 steps; 100 filter runs over 10 observation sequences (10 for OT and Gumbel).

	E_2				E_H (matched)				E_∞ (matched)			
Method	32	128	512	2048	32	128	512	2048	32	128	512	2048
DSR (τ=0.1, paper)	**5.12**	4.83	4.87	5.19	5.18	**4.82**	**4.84**	**4.87**	5.18	5.01	5.02	5.26
DSR (τ=0.1, unsorted)	4.83	4.95	4.84	4.91	**4.79**	**4.87**	**4.79**	**4.9**	4.8	5.01	4.9	5.07
DSR (τ=0.5)	5.14	4.81	4.87	4.9	**5.04**$^+$	**4.77**	**4.82**	**4.88**	5.28$^-$	4.94	4.96	5.05
DSR (τ=2.5/N)†	**5.11**	**5.04**	4.81	4.42	5.34	5.09	**4.74**	4.5	5.32	5.1	4.89	**4.41**
Systematic	5.08	**4.55**	4.13	4.14	5.35	4.59	4.11	4.16	**5.06**	4.62	**4.06**	**4.13**
Multinomial	5.38	4.79	4.21	4.16	**5.37**	4.92	**4.16**	**4.13**	5.42	**4.76**	4.19	4.22
Systematic, stop-gradient (PyDPF)	5.5	4.76	4.11	4.11	5.37	4.78	**4.09**	**4.09**	**5.26**	**4.61**	4.14	4.12
Soft (α=0.5)	5.62	4.8	**4.28**	4.2	**5.44**	**4.58**	4.34	**4.15**	5.8	5.03	4.31	4.24
OT (PyDPF)	5.31	5.65	–	–	5.92	5.76	–	–	**4.86**	**5.48**	–	–
OT (PFJAX)	4.97	5.56	–	–	5.71	5.68	–	–	**4.72**	**5.47**	–	–
OT (FilterFlow)	5.22	5.3	–	–	**4.99**	**5.24**	–	–	5.14	5.45	–	–
Eigen moment matching	5.12	**4.51**	4.21	4.24	**5.01**	4.53	4.21	**4.2**	5.19	4.61	**4.2**	4.28
Gaussian moment matching	5.23	4.51	4.36	4.2	5.28	**4.44**	4.29	4.21	**5.18**	4.56	**4.24**	**4.18**
Gumbel-softmax	**4.45**	**5.19**	–	–	4.53	5.28	–	–	4.63	5.42	–	–
Variational (KL)	**5.19**	4.66	4.2	4.27	5.28	4.61	4.16	**4.07**	5.21	**4.54**	**4.15**	4.16

†Local diagnostic, not from the paper: τ rescaled so that τN equals the paper's value at N=25. Cells are means; $^+/^-$: paired difference from E_2 on the same trials significantly better/worse (95% CI over observation sequences excludes 0). Bold marks the best trigger for that method and N. A dash means not run (dense methods stop at N=128).

Table 43: Bearing-only tracking: RMSE after the observation gap (lower is better), matched resampling rate. Resample when $E_H < \tau_H N$ or $E_\infty < \tau_\infty N$, with per-model thresholds calibrated so that systematic resampling resamples as often as under $E_2 < 0.5N$ (linear: τ_H=0.615, τ_∞=0.223; ungm: τ_H=0.567, τ_∞=0.351; volatility: τ_H=0.587, τ_∞=0.194; bearing: τ_H=0.596, τ_∞=0.285; lorenz63: τ_H=0.599, τ_∞=0.255; lorenz96: τ_H=0.62, τ_∞=0.187). 50 steps; 100 filter runs over 10 observation sequences (10 for OT and Gumbel).

	E_2				E_H (matched)				E_∞ (matched)			
Method	32	128	512	2048	32	128	512	2048	32	128	512	2048
DSR (τ=0.1, paper)	**9.69**	9.25	9.31	9.97	9.82	**9.21**	**9.26**	**9.31**	9.79	9.58	9.62	10.1
DSR (τ=0.1, unsorted)	9.15	9.52	9.33	9.46	9.08	**9.36**	**9.23**	**9.46**	**9.06**	9.61	9.35	9.68
DSR (τ=0.5)	9.77	9.22	9.38	9.44	**9.56**	**9.16**	**9.29**	**9.41**	10	9.41	9.45	9.61
DSR (τ=2.5/N)†	**9.66**	**9.53**	9.03	8.14	10.2	9.65	**8.89**	8.33	10.1	9.69	9.26	**8.14**
Systematic	**9.54**	**8.44**	7.61	7.58	10.2	8.54	7.55	7.67	9.55	8.67	**7.41**	**7.56**
Multinomial	10.1	8.92	7.76	7.65	**10.1**	9.17	**7.66**	**7.56**	10.2	**8.84**	7.68	7.77
Systematic, stop-gradient (PyDPF)	10.4	8.96	7.57	7.54	10.1	9.01	**7.51**	**7.5**	**9.95**	**8.53**	7.65	7.57
Soft (α=0.5)	10.6	9.01	**7.92**	7.71	**10.2**	**8.53**	8.11	**7.56**	11.1	9.34	8.02	7.83
OT (PyDPF)	9.98	10.9	–	–	11.4	11.3	–	–	**8.95**	**10.5**	–	–
OT (PFJAX)	9.19	10.7	–	–	11	11.1	–	–	**8.74**	**10.5**	–	–
OT (FilterFlow)	9.96	10.2	–	–	**9.48**	**10.1**	–	–	9.87	10.6	–	–
Eigen moment matching	9.59	**8.28**	7.59	7.69	**9.41**	8.34	7.58	**7.56**	9.81	8.54	**7.58**	7.76
Gaussian moment matching	9.74	8.3	7.98	7.62	9.92	**8.14**	7.76	7.6	**9.68**	8.47	**7.68**	**7.58**
Gumbel-softmax	**8.26**	**9.89**	–	–	8.53	10.1	–	–	8.69	10.5	–	–
Variational (KL)	**9.81**	8.76	7.73	7.9	9.99	8.63	7.65	**7.42**	9.85	**8.53**	**7.65**	7.62

†Local diagnostic, not from the paper: τ rescaled so that τN equals the paper's value at N=25. Cells are means; $^+/^-$: paired difference from E_2 on the same trials significantly better/worse (95% CI over observation sequences excludes 0). Bold marks the best trigger for that method and N. A dash means not run (dense methods stop at N=128).

Table 44: Bearing-only tracking: 95% coverage (closer to 0.95 is better), matched resampling rate. Resample when $E_H < \tau_H N$ or $E_\infty < \tau_\infty N$, with per-model thresholds calibrated so that systematic resampling resamples as often as under $E_2 < 0.5N$ (linear: τ_H=0.615, τ_∞=0.223; ungm: τ_H=0.567, τ_∞=0.351; volatility: τ_H=0.587, τ_∞=0.194; bearing: τ_H=0.596, τ_∞=0.285; lorenz63: τ_H=0.599, τ_∞=0.255; lorenz96: τ_H=0.62, τ_∞=0.187). 50 steps; 100 filter runs over 10 observation sequences (10 for OT and Gumbel).

	E_2				E_H (matched)				E_∞ (matched)			
Method	32	128	512	2048	32	128	512	2048	32	128	512	2048
DSR (τ=0.1, paper)	0.0692	0.0836	**0.0874**	0.0572	0.067	**0.0918**	0.0858	**0.0788**	**0.0704**	0.065	0.0538	0.0494
DSR (τ=0.1, unsorted)	0.0422	0.0202	0.0204	0.0174	0.0456	0.0222	**0.0222**	0.018	**0.0532**	**0.0268**	0.02	**0.0192**
DSR (τ=0.5)	0.02	0.0234	**0.0216**	**0.02**	0.0222	0.0228	0.021	**0.02**	**0.0252**	**0.0246**	0.0212	0.0192
DSR (τ=2.5/N)†	**0.0762**	0.299	0.494	0.682	0.0664	0.285	**0.508**	0.66	0.067	**0.302**	0.488	**0.689**
Systematic	**0.265**	0.636	0.826	0.904	0.242	0.634	0.825	0.904	0.259	**0.659**	**0.83**	**0.907**
Multinomial	**0.196**	0.54	0.809	**0.901**	0.195	**0.559**$^+$	**0.813**	0.892	**0.196**	0.551	0.802	0.893
Systematic, stop-gradient (PyDPF)	0.21	0.649	**0.85**	**0.904**	0.215	0.639	0.838	0.903	**0.23**	**0.675**	0.841	0.903
Soft (α=0.5)	**0.179**	0.519	**0.802**	0.891	0.159	**0.55**	0.782	**0.895**	0.15	0.521	0.787	0.887
OT (PyDPF)	0.09	0.04	–	–	0.05	0.04	–	–	**0.174**	**0.042**	–	–
OT (PFJAX)	0.092	**0.042**	–	–	0.048	**0.042**	–	–	**0.17**	**0.042**	–	–
OT (FilterFlow)	0.054	0.052	–	–	0.046	**0.064**	–	–	**0.074**	0.042	–	–
Eigen moment matching	0.51	**0.778**	0.87	**0.901**	**0.515**	0.768	0.871	0.895	0.484	**0.778**	**0.873**	0.881
Gaussian moment matching	0.362	0.724	0.85	**0.902**	**0.367**	0.73	0.86	0.896	0.352	**0.74**	**0.861**	0.89
Gumbel-softmax	0.196	0.216	–	–	**0.202**	**0.244**	–	–	0.198	0.22	–	–
Variational (KL)	**0.246**	0.633	0.83	0.873	0.231	0.613	0.836	**0.892**	0.235	**0.64**	**0.84**	0.883

†Local diagnostic, not from the paper: τ rescaled so that τN equals the paper's value at N=25. Cells are means; $^+/^-$: paired difference from E_2 on the same trials significantly better/worse (95% CI over observation sequences excludes 0). Bold marks the best trigger for that method and N. A dash means not run (dense methods stop at N=128).

Table 45: Bearing-only tracking: predictive NLL (lower is better), matched resampling rate. Resample when $E_H < \tau_H N$ or $E_\infty < \tau_\infty N$, with per-model thresholds calibrated so that systematic resampling resamples as often as under $E_2 < 0.5N$ (linear: τ_H=0.615, τ_∞=0.223; ungm: τ_H=0.567, τ_∞=0.351; volatility: τ_H=0.587, τ_∞=0.194; bearing: τ_H=0.596, τ_∞=0.285; lorenz63: τ_H=0.599, τ_∞=0.255; lorenz96: τ_H=0.62, τ_∞=0.187). 50 steps; 100 filter runs over 10 observation sequences (10 for OT and Gumbel).

	E_2				E_H (matched)				E_∞ (matched)			
Method	32	128	512	2048	32	128	512	2048	32	128	512	2048
DSR (τ=0.1, paper)	**−0.528**	−0.638	−0.612	−0.435	−0.472	**−0.651**	**−0.665**	**−0.569**	−0.512	-0.474^{-}	−0.423	−0.22
DSR (τ=0.1, unsorted)	−0.385	−0.536	−0.565	**−0.646**	**−0.422**	**−0.586**	**−0.596**	−0.599	−0.371	−0.457	-0.224^{-}	0.0573^{-}
DSR (τ=0.5)	0.104	−0.454	−0.545	−0.611	**−0.015**	**−0.49**	$\mathbf{-0.652}^{+}$	**−0.639**	0.351^{-}	-0.0942^{-}	−0.0816	0.107^{-}
DSR (τ=2.5/N)†	**−0.613**	−1.16	−1.27	−1.31	−0.586	**−1.17**	**−1.27**	−1.29	−0.591	−1.16	−1.24	**−1.31**
Systematic	**−1.2**	−1.31	−1.32	**−1.33**	−1.17	−1.31	−1.32	−1.33	−1.2	**−1.31**	**−1.32**	−1.33
Multinomial	−1.09	−1.3	**−1.32**	**−1.33**	**−1.1**	−1.29	−1.32	−1.33	−1.1	**−1.31**	−1.32	−1.33
Systematic, stop-gradient (PyDPF)	**−1.2**	−1.31	**−1.32**	**−1.33**	−1.18	−1.31	−1.32	−1.33	−1.19	**−1.31**	−1.32	−1.33
Soft (α=0.5)	**−1**	−1.29	−1.32	**−1.33**	0.0233	**−1.3**	**−1.32**	−1.33	−0.927	−1.3	−1.32	−1.33
OT (PyDPF)	−0.865	**−1.14**	–	–	−0.99	−1.09	–	–	**−1.09**	−0.807	–	–
OT (PFJAX)	−0.905	**−1.14**	–	–	**−0.991**	−1.08	–	–	−0.837	−0.813	–	–
OT (FilterFlow)	**−1.08**	−0.797	–	–	−0.904	−0.805	–	–	−0.846	**−0.977**	–	–
Eigen moment matching	**−1.27**	−1.32	−1.32	−1.32	−1.26	**−1.32**	**−1.32**	−1.32	−1.26	−1.32	−1.32	**−1.33**
Gaussian moment matching	−1.27	−1.32	−1.32	−1.33	−1.26	**−1.32**	−1.32	−1.33	**−1.28**	−1.32	**−1.33**	**−1.33**
Gumbel-softmax	**−1.21**	−1.23	–	–	−1.21	**−1.25**	–	–	−1.17	−1.23	–	–
Variational (KL)	**−1.2**	**−1.31**	−1.32	−1.32	−1.19	−1.29	−1.32	**−1.33**	−1.18	−1.3	**−1.32**	−1.32

†Local diagnostic, not from the paper: τ rescaled so that τN equals the paper's value at N=25. Cells are means; $^{+}/^{-}$: paired difference from E_2 on the same trials significantly better/worse (95% CI over observation sequences excludes 0). Bold marks the best trigger for that method and N. A dash means not run (dense methods stop at N=128).

Table 46: Bearing-only tracking: resampling steps out of 50, matched resampling rate. Resample when $E_H < \tau_H N$ or $E_\infty < \tau_\infty N$, with per-model thresholds calibrated so that systematic resampling resamples as often as under $E_2 < 0.5N$ (linear: τ_H=0.615, τ_∞=0.223; ungm: τ_H=0.567, τ_∞=0.351; volatility: τ_H=0.587, τ_∞=0.194; bearing: τ_H=0.596, τ_∞=0.285; lorenz63: τ_H=0.599, τ_∞=0.255; lorenz96: τ_H=0.62, τ_∞=0.187). 50 steps; 100 filter runs over 10 observation sequences (10 for OT and Gumbel).

	E_2				E_H (matched)				E_∞ (matched)			
Method	32	128	512	2048	32	128	512	2048	32	128	512	2048
DSR (τ=0.1, paper)	7.21	7.3	7.52	7.88	7.04	7.06	7.2	7.21	7.36	8.53	9.31	9.96
DSR (τ=0.1, unsorted)	7.19	6.81	6.57	6.39	7.01	6.47	6.3	6.21	7.27	7.73	8.62	9.32
DSR (τ=0.5)	6.64	6.4	6.34	6.39	6.27	6.07	5.88	5.97	7	7.66	8.49	9.11
DSR (τ=2.5/N)†	7.5	8.03	8.29	8.67	7.35	7.91	8.23	8.71	7.5	8.27	8.89	9.38
Systematic	8.85	9.35	9.48	9.35	9.05	9.41	9.7	9.59	8.27	9.27	9.57	9.93
Multinomial	9.03	9.39	9.44	9.5	9.05	9.58	9.6	9.62	8.67	9.1	9.59	9.99
Systematic, stop-gradient (PyDPF)	9	9.41	9.42	9.43	9.04	9.47	9.66	9.64	8.38	8.97	9.54	9.96
Soft (α=0.5)	13.5	14.1	14.3	14.4	14.7	15.2	15.6	15.7	11.7	12.7	13.2	13.5
OT (PyDPF)	8.4	7.9	–	–	7.5	7.4	–	–	7.4	8.7	–	–
OT (PFJAX)	8.3	7.9	–	–	7.6	7.4	–	–	7.6	8.5	–	–
OT (FilterFlow)	6.9	8.1	–	–	6.9	7.5	–	–	7.6	8.5	–	–
Eigen moment matching	9.48	9.51	9.57	9.45	9.62	9.8	9.75	9.78	9.02	9.42	9.94	10.3
Gaussian moment matching	9.72	9.61	9.55	9.51	9.86	9.81	9.76	9.88	8.72	9.43	9.72	10.3
Gumbel-softmax	7.7	8.4	–	–	7.5	8.2	–	–	8	9	–	–
Variational (KL)	8.8	9.28	9.33	9.35	8.88	9.28	9.51	9.5	8.19	9.18	9.5	9.87

†Local diagnostic, not from the paper: τ rescaled so that τN equals the paper's value at N=25. Cells are means; $^{+}/^{-}$: paired difference from E_2 on the same trials significantly better/worse (95% CI over observation sequences excludes 0). Bold marks the best trigger for that method and N. A dash means not run (dense methods stop at N=128).

Table 47: Stochastic Lorenz-63: state RMSE (lower is better), matched resampling rate. Resample when $E_H < \tau_H N$ or $E_\infty < \tau_\infty N$, with per-model thresholds calibrated so that systematic resampling resamples as often as under $E_2 < 0.5N$ (linear: τ_H=0.615, τ_∞=0.223; ungm: τ_H=0.567, τ_∞=0.351; volatility: τ_H=0.587, τ_∞=0.194; bearing: τ_H=0.596, τ_∞=0.285; lorenz63: τ_H=0.599, τ_∞=0.255; lorenz96: τ_H=0.62, τ_∞=0.187). 50 steps; 100 filter runs over 10 observation sequences (10 for OT and Gumbel).

	E_2				E_H (matched)				E_∞ (matched)			
Method	32	128	512	2048	32	128	512	2048	32	128	512	2048
DSR (τ=0.1, paper)	0.708	0.542	**0.491**	0.486	0.706	0.595	0.511	**0.484**	**0.614**	**0.533**	0.495	0.496^-
DSR (τ=0.1, unsorted)	0.853	**0.619**	**0.538**	**0.514**	1.24	0.727	0.563	0.518	**0.779**	0.639	0.549	0.533
DSR (τ=0.5)	**1.22**	0.792	**0.548**	**0.52**	1.62	0.742	0.598	**0.52**	1.29	**0.657**	0.559	0.535
DSR (τ=2.5/N)†	0.64	**0.479**	0.446	0.433	0.702	0.493	**0.444**	0.433	**0.591**	0.479	0.446	**0.433**
Systematic	0.563	0.456	0.437	0.431	0.559	**0.456**	**0.436**	0.432	**0.556**	0.461^-	0.437	$\mathbf{0.428}^+$
Multinomial	**0.549**	0.469	0.44	0.43	0.564	0.465	**0.439**	**0.429**	0.595	**0.463**	0.439	0.431
Systematic, stop-gradient (PyDPF)	0.584	0.464	0.439	0.431	**0.583**	0.464	**0.438**	0.431	0.586	**0.457**	0.438	**0.43**
Soft (α=0.5)	**0.616**	**0.467**	**0.44**	**0.43**	0.626	0.469	0.44	0.43	0.622	0.468	0.441	0.431
OT (PyDPF)	**0.542**	0.477	–	–	0.649	0.493	–	–	0.693	**0.475**	–	–
OT (PFJAX)	**0.55**	0.468	–	–	0.601	**0.467**	–	–	0.781	0.473	–	–
OT (FilterFlow)	**0.595**	0.491	–	–	0.652	0.491	–	–	0.813	**0.475**	–	–
Eigen moment matching	**0.533**	**0.456**	0.437	0.43	0.546	0.459	0.438	**0.43**	0.537	0.457	**0.437**	0.43
Gaussian moment matching	0.591	**0.46**	0.439	0.43	**0.577**	0.461	0.439	**0.43**	0.599	0.462	**0.439**	0.431
Gumbel-softmax	**0.756**	**0.484**	–	–	0.759	0.507	–	–	0.771	0.489	–	–
Variational (KL)	**0.546**	**0.461**	0.438	**0.429**	0.549	0.462	0.439	0.43	0.551	0.462	**0.437**	0.431

†Local diagnostic, not from the paper: τ rescaled so that τN equals the paper's value at N=25. Cells are means; $^+/^-$: paired difference from E_2 on the same trials significantly better/worse (95% CI over observation sequences excludes 0). Bold marks the best trigger for that method and N. A dash means not run (dense methods stop at N=128).

Table 48: Stochastic Lorenz-63: 95% coverage (closer to 0.95 is better), matched resampling rate. Resample when $E_H < \tau_H N$ or $E_\infty < \tau_\infty N$, with per-model thresholds calibrated so that systematic resampling resamples as often as under $E_2 < 0.5N$ (linear: τ_H=0.615, τ_∞=0.223; ungm: τ_H=0.567, τ_∞=0.351; volatility: τ_H=0.587, τ_∞=0.194; bearing: τ_H=0.596, τ_∞=0.285; lorenz63: τ_H=0.599, τ_∞=0.255; lorenz96: τ_H=0.62, τ_∞=0.187). 50 steps; 100 filter runs over 10 observation sequences (10 for OT and Gumbel).

	E_2				E_H (matched)				E_∞ (matched)			
Method	32	128	512	2048	32	128	512	2048	32	128	512	2048
DSR (τ=0.1, paper)	0.397	0.579	**0.655**	0.682	0.401	**0.58**	0.649	**0.685**	$\mathbf{0.416}^+$	0.577	0.636^-	0.653^-
DSR (τ=0.1, unsorted)	0.378	0.516	0.576	0.597	0.378	**0.521**	**0.58**	**0.604**	**0.384**	0.501^-	0.563	0.578
DSR (τ=0.5)	0.332	0.489	0.563	0.593	**0.334**	**0.495**	**0.568**	**0.599**	0.333	0.476	0.549	0.576
DSR (τ=2.5/N)†	0.418	**0.723**	0.877	**0.932**	0.419	0.716	**0.88**	0.929	$\mathbf{0.433}^+$	0.719	0.875	$\mathbf{0.932}^+$
Systematic	0.58	0.854	0.933	**0.95**	**0.584**	**0.856**	**0.936**	0.949	0.581	0.847	0.932	0.952
Multinomial	**0.581**	0.833	**0.928**	**0.95**	0.564	**0.836**	**0.928**	0.951	0.566	0.835	0.927	**0.95**
Systematic, stop-gradient (PyDPF)	0.568	0.849	0.928	**0.95**	**0.574**	0.849	**0.928**	0.95	0.566	**0.852**	0.927	0.951
Soft (α=0.5)	0.473	0.814	0.922	**0.95**	0.473	0.812	**0.922**	0.95	**0.475**	**0.815**	**0.922**	0.949
OT (PyDPF)	**0.506**	0.772	–	–	0.49	0.762	–	–	0.484	**0.79**	–	–
OT (PFJAX)	0.532	**0.804**	–	–	**0.542**	0.8	–	–	0.486	**0.804**	–	–
OT (FilterFlow)	0.484	0.734	–	–	**0.498**	0.744	–	–	0.496	**0.748**	–	–
Eigen moment matching	**0.612**	**0.858**	**0.932**	0.949	0.609	0.855	0.931	**0.95**	0.598	**0.858**	0.932	0.951
Gaussian moment matching	0.568	**0.847**	**0.929**	0.947	**0.57**	0.847	0.928	**0.948**	0.557	0.846	0.929	0.947
Gumbel-softmax	0.464	**0.702**	–	–	0.456	0.686	–	–	**0.476**	0.688	–	–
Variational (KL)	**0.588**	0.845	**0.927**	**0.949**	0.584	**0.846**	0.927	0.946	0.576	0.846	0.925	**0.951**

†Local diagnostic, not from the paper: τ rescaled so that τN equals the paper's value at N=25. Cells are means; $^+/^-$: paired difference from E_2 on the same trials significantly better/worse (95% CI over observation sequences excludes 0). Bold marks the best trigger for that method and N. A dash means not run (dense methods stop at N=128).

Table 49: Stochastic Lorenz-63: predictive NLL (lower is better), matched resampling rate. Resample when $E_H < \tau_H N$ or $E_\infty < \tau_\infty N$, with per-model thresholds calibrated so that systematic resampling resamples as often as under $E_2 < 0.5N$ (linear: τ_H=0.615, τ_∞=0.223; ungm: τ_H=0.567, τ_∞=0.351; volatility: τ_H=0.587, τ_∞=0.194; bearing: τ_H=0.596, τ_∞=0.285; lorenz63: τ_H=0.599, τ_∞=0.255; lorenz96: τ_H=0.62, τ_∞=0.187). 50 steps; 100 filter runs over 10 observation sequences (10 for OT and Gumbel).

	E_2				E_H (matched)				E_∞ (matched)			
Method	32	128	512	2048	32	128	512	2048	32	128	512	2048
DSR (τ=0.1, paper)	35.3	31.8	**31.4**	**31.4**	35.2	33.3	31.6	31.4	**32.9**	**31.7**	31.4	31.4$^-$
DSR (τ=0.1, unsorted)	38.4	33.4	**31.8**	**31.6**	47.3	36.3	32.5	31.6	**36.9**	**33.4**	32	31.7
DSR (τ=0.5)	**46**	36.6	**31.9**	**31.6**	57.9	36.6	33.4	31.6	46.6	**33.6**	32	31.7
DSR (τ=2.5/N)†	33.7	31.3	31.2	31.2	35.4	31.5	**31.2**	31.2	**32.3**	**31.3**	31.2	**31.2**
Systematic	31.8	31.2	**31.2**	31.2	31.8	**31.2**$^+$	31.2	31.2	**31.7**	31.2	31.2	**31.2**$^+$
Multinomial	**31.6**	**31.2**	**31.2**	31.2	31.7	31.2	31.2	**31.2**	32.3	31.2	**31.2**	31.2
Systematic, stop-gradient (PyDPF)	32.2	31.2	31.2	31.2	32.2$^+$	31.2	**31.2**	31.2	**32.1**	**31.2**$^+$	31.2	**31.2**
Soft (α=0.5)	**32.2**	31.2	**31.2**	**31.2**	32.3	31.3	31.2	**31.2**	32.4	**31.2**	31.2	31.2
OT (PyDPF)	**31.5**	31.3	–	–	32.8	31.4	–	–	33.8	**31.3**	–	–
OT (PFJAX)	**31.5**	31.3	–	–	32.1	31.3	–	–	36.9	**31.3**	–	–
OT (FilterFlow)	**31.7**	31.4	–	–	32.4	31.5	–	–	37.8	**31.4**	–	–
Eigen moment matching	**31.5**	**31.2**	**31.2**	**31.2**	31.7	31.2	31.2	**31.2**	31.6	31.2	31.2	31.2
Gaussian moment matching	32.3	**31.2**	**31.2**	31.2	**32.1**	31.2	**31.2**	**31.2**$^+$	32.4	31.2	31.2	31.2
Gumbel-softmax	**35.4**	**31.3**	–	–	**35.4**	31.5	–	–	36.3	31.4	–	–
Variational (KL)	**31.6**	31.2	**31.2**	**31.2**	31.6	**31.2**	31.2	**31.2**	31.7	31.2	31.2	31.2

†Local diagnostic, not from the paper: τ rescaled so that τN equals the paper's value at N=25. Cells are means; $^+/^-$: paired difference from E_2 on the same trials significantly better/worse (95% CI over observation sequences excludes 0). Bold marks the best trigger for that method and N. A dash means not run (dense methods stop at N=128).

Table 50: Stochastic Lorenz-63: resampling steps out of 50, matched resampling rate. Resample when $E_H < \tau_H N$ or $E_\infty < \tau_\infty N$, with per-model thresholds calibrated so that systematic resampling resamples as often as under $E_2 < 0.5N$ (linear: τ_H=0.615, τ_∞=0.223; ungm: τ_H=0.567, τ_∞=0.351; volatility: τ_H=0.587, τ_∞=0.194; bearing: τ_H=0.596, τ_∞=0.285; lorenz63: τ_H=0.599, τ_∞=0.255; lorenz96: τ_H=0.62, τ_∞=0.187). 50 steps; 100 filter runs over 10 observation sequences (10 for OT and Gumbel).

	E_2				E_H (matched)				E_∞ (matched)			
Method	32	128	512	2048	32	128	512	2048	32	128	512	2048
DSR (τ=0.1, paper)	38.9	39	38.9	38.7	38.5	38.6	38.6	38.2	36.9	39.1	40.4	41.3
DSR (τ=0.1, unsorted)	38.8	38.4	38.1	38	38.5	38	37.6	37.3	37.2	39.2	40.1	40.8
DSR (τ=0.5)	38.8	38.6	38.1	38.1	38.5	37.9	37.5	37.5	37.3	39.2	40.1	40.8
DSR (τ=2.5/N)†	39	40.3	41.4	42.4	38.8	40	41.2	42.1	36.9	40.2	42.3	42.8
Systematic	41.6	42.9	43.3	43.3	41.4	42.5	43	43.1	39	42.1	43.3	43.5
Multinomial	41.5	42.7	43.3	43.2	41.4	42.5	43.1	43.1	38.7	42	43.3	43.5
Systematic, stop-gradient (PyDPF)	41.4	43	43.1	43.3	41.1	42.6	43	43.1	38.9	42.1	43.1	43.5
Soft (α=0.5)	48.4	49.1	49.1	49	48.7	49.2	49.3	49.2	46.2	48.6	49	49.2
OT (PyDPF)	41.3	41	–	–	42.1	41.2	–	–	37.7	41.4	–	–
OT (PFJAX)	40.1	41.2	–	–	40.5	40.9	–	–	37.5	41.4	–	–
OT (FilterFlow)	40	40.7	–	–	40.2	40.1	–	–	38.4	40	–	–
Eigen moment matching	41.4	42.9	43.2	43.2	41.5	42.6	43.1	43.1	38.8	42	43.2	43.3
Gaussian moment matching	41.3	42.7	43.3	43.2	41.2	42.5	43.2	43.1	38.6	42.2	43.3	43.4
Gumbel-softmax	39.9	40.1	–	–	39.4	39.7	–	–	37	39.7	–	–
Variational (KL)	41.3	42.7	43.2	43.1	41.2	42.4	42.9	43	38.7	42	43.2	43.4

†Local diagnostic, not from the paper: τ rescaled so that τN equals the paper's value at N=25. Cells are means; $^+/^-$: paired difference from E_2 on the same trials significantly better/worse (95% CI over observation sequences excludes 0). Bold marks the best trigger for that method and N. A dash means not run (dense methods stop at N=128).

Table 51: Stochastic Lorenz-96: state RMSE (lower is better), matched resampling rate. Resample when $E_H < \tau_H N$ or $E_\infty < \tau_\infty N$, with per-model thresholds calibrated so that systematic resampling resamples as often as under $E_2 < 0.5N$ (linear: τ_H=0.615, τ_∞=0.223; ungm: τ_H=0.567, τ_∞=0.351; volatility: τ_H=0.587, τ_∞=0.194; bearing: τ_H=0.596, τ_∞=0.285; lorenz63: τ_H=0.599, τ_∞=0.255; lorenz96: τ_H=0.62, τ_∞=0.187). 50 steps; 100 filter runs over 10 observation sequences (10 for OT and Gumbel).

	E_2				E_H (matched)				E_∞ (matched)			
Method	32	128	512	2048	32	128	512	2048	32	128	512	2048
DSR (τ=0.1, paper)	3.09	2.46	2.11	1.9	3.09	**2.46**	**2.11**$^+$	**1.88**$^+$	**3.08**	2.46	2.12$^-$	1.91
DSR (τ=0.1, unsorted)	3.07	2.47	2.13	1.9	3.07	**2.46**$^+$	**2.11**$^+$	**1.88**$^+$	**3.05**	2.47	2.13	1.91$^-$
DSR (τ=0.5)	3.2	2.65	2.25	1.97	3.19	**2.64**	**2.24**$^+$	**1.96**$^+$	**3.18**	2.64	2.25$^-$	1.99$^-$
DSR (τ=2.5/N)†	3.02	2.17	**1.45**	1.08	3.02	**2.16**	1.45	**1.08**	**3**	2.16	1.45	1.08
Systematic	2.76	2.09	1.45	1.07	2.76	**2.09**	**1.45**	**1.07**	**2.74**	2.09	1.45	1.07
Multinomial	2.8	2.06	**1.41**	1.09	**2.8**	2.06	1.41	1.09	2.82	**2.06**	1.41	**1.09**
Systematic, stop-gradient (PyDPF)	2.73	**2.05**	1.41	**1.06**	**2.73**	2.05	1.41	**1.06**	2.74	2.05	**1.41**	1.06
Soft (α=0.5)	**2.96**	2.14	**1.48**	**1.07**	**2.96**	2.14	**1.48**	**1.07**	2.97	**2.14**	**1.48**	**1.07**
OT (PyDPF)	3.11	2.44	–	–	**3.11**	**2.44**	–	–	3.16	2.44	–	–
OT (PFJAX)	**3.27**	2.28	–	–	3.27	2.28	–	–	3.27	**2.28**	–	–
OT (FilterFlow)	3.36	**2.52**	–	–	3.36	2.52	–	–	**3.21**	2.52	–	–
Eigen moment matching	**2.62**	1.75	1.1	0.849	2.62	**1.75**	**1.1**	**0.849**	2.62	1.75	1.1	0.849
Gaussian moment matching	**2.55**	**1.8**	**1.15**	**0.865**	2.55	1.8	1.15	0.865	2.56	**1.8**	1.15	**0.865**
Gumbel-softmax	3.15	2.44	–	–	3.15	**2.43**	–	–	**3.14**	2.43	–	–
Variational (KL)	**2.77**	2.05	1.41	1.09	2.77	2.05	**1.41**	1.09	2.8$^-$	**2.05**	1.41	**1.09**

†Local diagnostic, not from the paper: τ rescaled so that τN equals the paper's value at N=25. Cells are means; $^+/^-$: paired difference from E_2 on the same trials significantly better/worse (95% CI over observation sequences excludes 0). Bold marks the best trigger for that method and N. A dash means not run (dense methods stop at N=128).

Table 52: Stochastic Lorenz-96: 95% coverage (closer to 0.95 is better), matched resampling rate. Resample when $E_H < \tau_H N$ or $E_\infty < \tau_\infty N$, with per-model thresholds calibrated so that systematic resampling resamples as often as under $E_2 < 0.5N$ (linear: τ_H=0.615, τ_∞=0.223; ungm: τ_H=0.567, τ_∞=0.351; volatility: τ_H=0.587, τ_∞=0.194; bearing: τ_H=0.596, τ_∞=0.285; lorenz63: τ_H=0.599, τ_∞=0.255; lorenz96: τ_H=0.62, τ_∞=0.187). 50 steps; 100 filter runs over 10 observation sequences (10 for OT and Gumbel).

	E_2				E_H (matched)				E_∞ (matched)			
Method	32	128	512	2048	32	128	512	2048	32	128	512	2048
DSR (τ=0.1, paper)	**0**	**0.0024**	0.004	0.0102	**0**	0.002	**0.0082**	**0.016**$^+$	**0**	0.0018	0.002	0.0026
DSR (τ=0.1, unsorted)	0	0.0016	0.0024	0.006	0	**0.0024**	**0.0052**$^+$	**0.0124**	$\mathbf{2\times10^{-4}}$	0.0014	0.0016	8×10^{-4}
DSR (τ=0.5)	**0**	4×10^{-4}	0.001	0.0054	**0**	**0.0012**	**0.0046**	**0.0104**	**0**	4×10^{-4}	4×10^{-4}	8×10^{-4}
DSR (τ=2.5/N)†	2×10^{-4}	0.008	0.0828	0.272	0	**0.0102**	**0.0944**$^+$	**0.284**$^+$	$\mathbf{4\times10^{-4}}$	0.0076	0.0756	0.275
Systematic	**0.0016**	0.0788	0.303	0.563	0.0014	**0.0794**	**0.305**	**0.563**	**0.0016**	**0.0794**	0.301	0.563
Multinomial	0.001	0.0634	**0.308**	0.544	**0.0014**	0.0638	**0.308**	**0.545**	0	**0.0668**	0.306	0.544
Systematic, stop-gradient (PyDPF)	**0.001**	**0.0852**	0.321	**0.56**	**0.001**	0.0808	0.319	0.559	6×10^{-4}	0.0812	**0.322**	0.56
Soft (α=0.5)	**0.0012**	0.035	**0.249**	**0.519**	**0.0012**	0.035	**0.249**	**0.519**	6×10^{-4}	**0.0354**	**0.249**	**0.519**
OT (PyDPF)	**0**	**0.004**	–	–	**0**	**0.004**	–	–	**0**	**0.004**	–	–
OT (PFJAX)	**0**	0.034	–	–	**0**	**0.038**	–	–	**0**	0.036	–	–
OT (FilterFlow)	**0**	**0.002**	–	–	**0**	0	–	–	**0**	0	–	–
Eigen moment matching	**0.0052**	0.156	0.514	0.713	0.004	**0.158**	0.514	0.713	0.0036	0.154	**0.514**	**0.714**
Gaussian moment matching	0.0032	0.147	**0.48**	**0.706**	**0.0044**	0.147	**0.48**	0.705	0.0024	**0.148**	0.479	**0.706**
Gumbel-softmax	**0.002**	0.016	–	–	**0.002**	0.014	–	–	**0.002**	**0.018**	–	–
Variational (KL)	0.0022	**0.0704**	0.323	0.543	**0.0024**	0.0662	0.323	**0.544**	0.0016	0.0668	**0.324**	0.544

†Local diagnostic, not from the paper: τ rescaled so that τN equals the paper's value at N=25. Cells are means; $^+/^-$: paired difference from E_2 on the same trials significantly better/worse (95% CI over observation sequences excludes 0). Bold marks the best trigger for that method and N. A dash means not run (dense methods stop at N=128).

Table 53: Stochastic Lorenz-96: predictive NLL (lower is better), matched resampling rate. Resample when $E_H < \tau_H N$ or $E_\infty < \tau_\infty N$, with per-model thresholds calibrated so that systematic resampling resamples as often as under $E_2 < 0.5N$ (linear: τ_H=0.615, τ_∞=0.223; ungm: τ_H=0.567, τ_∞=0.351; volatility: τ_H=0.587, τ_∞=0.194; bearing: τ_H=0.596, τ_∞=0.285; lorenz63: τ_H=0.599, τ_∞=0.255; lorenz96: τ_H=0.62, τ_∞=0.187). 50 steps; 100 filter runs over 10 observation sequences (10 for OT and Gumbel).

	E_2				E_H (matched)				E_∞ (matched)			
Method	32	128	512	2048	32	128	512	2048	32	128	512	2048
DSR (τ=0.1, paper)	69.2	51.5	42.7	37.8	69.3	**51.4**	**42.6**$^+$	**37.6**$^+$	**69**	51.5	42.8$^-$	37.9
DSR (τ=0.1, unsorted)	68	52	43	38	68	**51.9**$^+$	**42.8**$^+$	**37.7**$^+$	**67.6**	52	43.1	38.2$^-$
DSR (τ=0.5)	72.1	56.5	45.8	39	71.8	**56.3**	**45.6**$^+$	**38.8**$^+$	**71.6**	56.4	45.9$^-$	39.2$^-$
DSR (τ=2.5/N)†	67.1	45	**29.4**	22.8	67.2	**44.9**	29.4	**22.8**$^+$	**66.8**	45	29.4	22.8
Systematic	60	**43.8**	30.1	**22.7**	60	43.8	**30.1**	22.7	**59.8**	**43.8**	30.1	22.7
Multinomial	60.9	43.1	**29.3**	**23.1**	**60.9**	43.1	29.3	23.1	61.6	**43**$^+$	29.3	**23.1**
Systematic, stop-gradient (PyDPF)	59.3	**42.9**	**29**	22.6	**59.3**	42.9	29	22.6	59.4	42.9	**29**	**22.6**
Soft (α=0.5)	**65.7**	**44.9**	**30.2**	**22.4**	**65.7**	**44.9**	**30.2**	**22.4**	65.9	44.9	**30.2**	**22.4**
OT (PyDPF)	68.1	54.7	–	–	**68.1**	**54.7**	–	–	69.7	54.8	–	–
OT (PFJAX)	**73.2**	49.5	–	–	**73.2**	49.5	–	–	73.4	**49.5**	–	–
OT (FilterFlow)	76.2	**56.4**	–	–	76.2	56.4	–	–	**70.6**	56.4	–	–
Eigen moment matching	57	36	23.2	**19.4**	**57**	**36**	**23.2**	**19.4**	57.2	36	23.2	**19.4**
Gaussian moment matching	**54**	37	**24.3**	**19.9**	54.1	37	24.3	**19.9**	54.7	**37**	24.3	19.9
Gumbel-softmax	70.5	55.1	–	–	70.5	**55**	–	–	**70.3**	55.1	–	–
Variational (KL)	**60.2**	42.7	29	23.2	60.2	42.7	29	23.2	60.8	**42.7**	**29**	**23.2**

†Local diagnostic, not from the paper: τ rescaled so that τN equals the paper's value at N=25. Cells are means; $^+/^-$: paired difference from E_2 on the same trials significantly better/worse (95% CI over observation sequences excludes 0). Bold marks the best trigger for that method and N. A dash means not run (dense methods stop at N=128).

Table 54: Stochastic Lorenz-96: resampling steps out of 50, matched resampling rate. Resample when $E_H < \tau_H N$ or $E_\infty < \tau_\infty N$, with per-model thresholds calibrated so that systematic resampling resamples as often as under $E_2 < 0.5N$ (linear: τ_H=0.615, τ_∞=0.223; ungm: τ_H=0.567, τ_∞=0.351; volatility: τ_H=0.587, τ_∞=0.194; bearing: τ_H=0.596, τ_∞=0.285; lorenz63: τ_H=0.599, τ_∞=0.255; lorenz96: τ_H=0.62, τ_∞=0.187). 50 steps; 100 filter runs over 10 observation sequences (10 for OT and Gumbel).

	E_2				E_H (matched)				E_∞ (matched)			
Method	32	128	512	2048	32	128	512	2048	32	128	512	2048
DSR (τ=0.1, paper)	49.3	49.1	49.1	48.8	48.9	48.5	48.1	47.5	47.7	49.3	49.8	49.9
DSR (τ=0.1, unsorted)	49.4	49.3	49.2	49	49.1	48.4	48.1	47.5	47.9	49.3	49.9	50
DSR (τ=0.5)	49.4	49.4	49.3	49	49	48.6	48.4	47.9	48.2	49.4	49.9	50
DSR (τ=2.5/N)†	49.1	49.2	49.1	49.3	48.8	48.5	48.2	48.6	47.5	49.3	49.6	49.9
Systematic	49.6	49.8	49.8	49.9	49.4	49.7	49.7	49.8	47.9	49.7	49.9	50
Multinomial	49.6	49.8	49.9	50	49.4	49.7	49.8	49.9	48.1	49.7	49.9	50
Systematic, stop-gradient (PyDPF)	49.6	49.8	49.9	49.9	49.4	49.6	49.8	49.8	47.8	49.6	49.9	50
Soft (α=0.5)	50	50	50	50	50	50	50	50	49.4	50	50	50
OT (PyDPF)	49.7	49.6	–	–	49.5	49.1	–	–	48.4	49.6	–	–
OT (PFJAX)	49.9	49.5	–	–	49.5	49.1	–	–	48.5	49.2	–	–
OT (FilterFlow)	49.6	49.5	–	–	49.5	49.1	–	–	48.4	49.4	–	–
Eigen moment matching	49.5	49.8	49.9	49.9	49.5	49.8	49.8	49.8	47.8	49.8	50	50
Gaussian moment matching	49.6	49.8	49.9	49.9	49.4	49.7	49.8	49.8	47.6	49.7	50	50
Gumbel-softmax	49.4	49.5	–	–	49.2	49	–	–	48.4	49.4	–	–
Variational (KL)	49.6	49.8	49.8	49.9	49.4	49.7	49.7	49.8	47.9	49.7	49.9	50

†Local diagnostic, not from the paper: τ rescaled so that τN equals the paper's value at N=25. Cells are means; $^+/^-$: paired difference from E_2 on the same trials significantly better/worse (95% CI over observation sequences excludes 0). Bold marks the best trigger for that method and N. A dash means not run (dense methods stop at N=128).

Conclusions: resampling triggers

Setup. Each trigger was run through the same harness, observation sequences, propagation noise and resampler seeds as the published $E_2 < 0.5N$ runs, so every trial is compared with its own E_2 counterpart. The comparison covers 6 filtering models, $N \in \{32, 128, 512, 2048\}$ ($N \leq 128$ for OT and Gumbel) and 15 resamplers. The common-threshold and matched-rate sets add 672 cells each, 1344 in total; none failed, and no trial had a numerical failure. A cell counts as *better* or *worse* when the 95% CI of the paired difference, clustered by observation sequence, excludes 0.

How often each trigger resamples. At the common threshold $0.5N$ the measures are ordered $E_\infty \leq E_2 \leq E_H$, so they fire at different rates. Relative to E_2, E_H resamples 4–29% less often and E_∞ 1.0–3.0× as often; the largest change is on stochastic volatility and the smallest on Lorenz-96, which resamples at nearly every step under every trigger. $CV^2 > c$ adds no new rule: $CV^2 = N/E_2 - 1$, so $CV^2 > 1$ is exactly $E_2 < 0.5N$, and its trials are identical to the baseline.

Standard resamplers are insensitive to the trigger. The standard methods are systematic, multinomial, stop-gradient, soft, OT, eigen and Gaussian moment matching, Gumbel-softmax and variational KL. For these methods, only 4.4% of (cell, metric) comparisons are significant. That is the false-positive rate expected at 95% confidence, and the significant cells are scattered with no consistent direction: each method ties E_2 in 20–24 of its 24 cells (11–12 of 12 for OT and Gumbel) on RMSE, predictive NLL and coverage. Within this range of resampling rates, the choice among E_2, E_H and E_∞ does not change their accuracy.

DSR is sensitive, and less resampling helps. For the four DSR variants, 20% of comparisons are significant, and their direction is consistent.

- *Resampling more often (E_∞) hurts DSR.* Bearing-only tracking: predictive NLL worsens by 0.49–1.36 nats, significant in 13 of 16 DSR cells, and paper-DSR coverage falls from 0.087 to 0.022 at N=512. Lorenz-63: coverage drops 3–11 points, significant in 15 of 16 cells. Lorenz-96: RMSE and NLL become slightly worse ($< 1\%$).
- *Resampling less often (E_H) helps DSR modestly.* Lorenz-96: RMSE improves by 0.7–4.4% and NLL by up to 1.07 nats, each significant in 11 of 16 cells. Bearing at N=512: paper-DSR RMSE improves from 4.87 to 4.67. Bearing NLL also improves for the unsorted and τ=0.5 variants.
- *Rescaled DSR (τ=2.5/N) is much less sensitive.* The exception is bearing coverage, where E_∞ still costs 8–12 points at $N \geq 128$.

This fits the static and gradient results. Each DSR call contracts the particle cloud, so the error is paid per resampling step, and a trigger that fires more often accumulates more of it. For DSR the trigger behaves like a tuning parameter; for unbiased resamplers it does not.

At a matched resampling rate, the measure barely matters. To separate the measure from the rate, the E_H and E_∞ thresholds were recalibrated per model so that systematic resampling resamples as often as under $E_2 < 0.5N$ ($\tau_H = 0.57$–0.62, $\tau_\infty = 0.19$–0.35). For the standard resamplers, rates then stay within 6% of E_2 in every model except stochastic volatility under E_H (11% fewer resamplings). They tie E_2 in 97–98% of comparisons, with significant cells split evenly between better and worse, which is a null result. For DSR the share of significant comparisons falls from 18–23% to 8–9%, and the effects shrink. On Lorenz-63, the 15 of 16 coverage losses under E_∞ become 3 better and 3 worse. On Lorenz-96, the E_H RMSE gain falls from 0.7–4.4% to 0.2–1.2%. The differences that remain follow DSR's own rate, which the systematic-based calibration does not match exactly. On bearing, DSR under matched E_∞ still resamples 17–46% more often than under E_2, and the worst NLL cells (0.34–0.72 nats) are the ones with the largest mismatch. On Lorenz-96, matched E_H resamples 1–2% less and gives small gains, while matched E_∞ resamples 1–2% more and gives small losses. We therefore find no evidence that any degeneracy measure is better than another at equal rate. For DSR, the resampling rate is what matters. A side effect is that a threshold calibrated on one resampler does not transfer to DSR: E_∞ fires more often on DSR's contracted clouds than on systematic ones.

What the trigger does not fix. No trigger closes the gap between paper-DSR and unbiased resampling. On bearing at N=512, its best coverage (0.113 under E_H) remains far from systematic resampling (0.826).

Caveats. The matched rates are calibrated on systematic resampling with 30 seeds at $N \in \{128, 512\}$, so for DSR they are only approximately matched; a calibration on each method's own rate would test the measure more cleanly. On Lorenz-96, coverage is near 0 under every trigger, so the "better" coverage verdicts there are changes of about 0.01. The results cover 50-step filters on these six models only. E-MIM is not included, because its exact definition was not available.